



UNDP/GEF Project (Steel Rerolling Mills)
Ministry of Steel, Government of India

PROJECT MANAGEMENT CELL, NEW DELHI

BASE DOCUMENT

STANDARD OPERATING PRACTICES

RE-ROLLING MILL PROPER



Compiled and Prepared

BY

SYCOM PROJECTS CONSULTANTS PVT. LTD., NEW DELHI

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SUBMITTED TO



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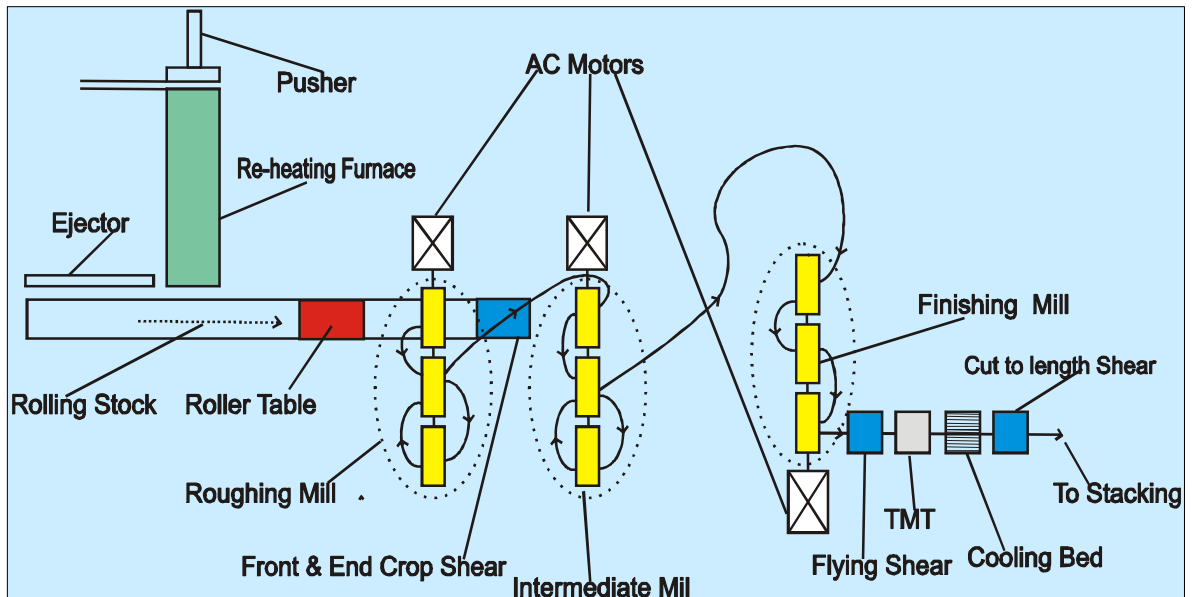
CHAPTER 1

DESCRIPTION OF STANDARD ROLLING MILL, MAJOR EQUIPMENTS/PARTS

1.1 MAJOR TYPES OF ROLLING MILLS

- The most common type of Steel Re-Rolling Mill in the SME Sector in India is the cross country Mill. The other types include semi continuous mills which are a combination of the cross country and the continuous Rolling Mills and the continuous Rolling Mills. The continuous Rolling Mills are virtually non-existent in the SME sector.
- A typical layout of a Cross Country Type Rolling Mill with 3 Main Drive AC Motors, one each for Roughing, Intermediate & Finishing Mills and 3 Stands in Roughing Mill; 3 Stands in Intermediate Mill & 3 stands in Finishing Mill is provided at **Figure 1.1**.

Figure 1.1: Typical Layout of 3-Drive Cross Country Type Rolling Mill



1.2 LIST OF MAJOR EQUIPMENTS/PARTS OF STANDARD CROSS COUNTRY TYPE ROLLING MILL

➤ A standard Cross Country type Rolling Mill in the SME-SRRM sector has the following major Equipments/Parts:

1. **Roughing Mill**: A Group Of 3/4 Stands (3 Hi) where maximum reduction of the Hot Bar takes place.
2. **Intermediate Mill**: A group of 2/3 stands (3 Hi/2 Hi) where the shaping of the product starts.
3. **Finishing Mill**: A group of 2/3 stands (3 Hi/2 Hi) where the final dimensions and shape of the product is achieved.
4. **Mill Main Drive Motors** : These are heavy duty (450-1350 HP) AC Motors of squirrel cage slip ring induction type which drive the roughing, intermediate & finishing mills. In addition DC motors of approx.250HP capacity are used to drive the continuous stand.
5. **Reduction Gear Box**: This is attached to the main mill AC drive motor and reduces the speed typically in the ratio of 1:6.
6. **Pinion Gear Box** : This is attached to the reduction gear box and has one input shaft and three output shafts to drive 3 Hi Mill Rolls.
7. **Mill Rolls**: These are loaded on the mill stands i.e. 3 rolls (bottom, middle, top) for 3 Hi Mill Stand, rotate and are used to reduce the hot bar dimensions, provide desired shape and size.
8. **Roll Neck Bearings** : Bearings either fiber or antifricition type are provided on the neck of the rolls to absorb shock and provide cushioning effect.
9. **Spindles & Couplings** :These are used to transmit power from the motor to the mill stands.

10. Repeaters: It is a guide that guides the bar exiting from a stand to the succeeding stand into the correct roll pass, without manual intervention.

11. Guides, Guards & Strippers:

- Are mounted on rest bars of each stand to guide the bar entering or exiting from the rolls.
- These guides could either be open/closed friction guides or roller guides on the ingoing side and there are side guards & stripper guides on the exit side to keep the rolled material from going off into a tangent.
- In the friction type of guides, the rolling stock is held by the friction inserts. These inserts wear out faster and the holding of the rolling stocks becomes ineffective resulting in either improper feeding of material or bad quality of the product. The roller guides are improved version of friction guides and the material is guided through rolls which reduces friction, wear & tear of guides.
- The strippers mounted on the guard assembly strip the bar exiting from the roll pass and prevents it from collaring or wrapping around the roll. The strippers also prevent the propagation of splits at the exit of the roll pass.

12. Roller Tables : Roller tables are used for automatic movement of hot & cold bar in various sections of the SRRM. These include RHF discharge roller table, roughing mill feeding roller table, Roller table for discharge of the bar from the finishing mill to the cooling bed, discharge of finished cold bar from the cooling bed etc. These roller tables consist of a number of steel rolls that are interconnected through pulleys & V belts and are generally driven by single AC motor of approx 15 HP capacity.

13. Tilting or Lifting Tables : In 3 Hi mills, the stock has to be mechanically lifted from the pass line of the middle & bottom rolls to the higher pass line of the middle & top rolls and to achieve this tilting tables on either or both sides of the stands may be used. These tables are recommended for mills where sections (profiles) are rolled.

14. Y-Roller Tables : Y -Roller Tables are used in cross country mills for automatically transferring stock from one stand to another. Y-roller tables are preferred for bar & rod mills where the input stock weight is upto 150kg.

15. Front & Back Ends Cutting Shears

- These are swivel type rotary shears used for cutting the front and back ends of the hot bars leaving the roughing mill and before entry into the intermediate stands.

16. Flying Shear

- The front end of the Bar leaving the Finishing Mill is cut before it enters the TMT box. These shears are provided in mills producing TMT bars.

17. TMT (THERMO MECHANICAL TREATMENT) SYSTEM

- The cut lengths then enter the TMT box in which high pressure (5- 7.5kg/cm²) water is sprayed on the bar to rapid water quench it for martempering it to achieve the maximum strength 500 – 580 N/mm² (Fe 500 – Fe 580).

18. Cooling Bed

- The Cooling Bed is generally W-Channel type where the material movement takes place manually with the help of tongs.

19. Cut to Length Shear: This is a fixed type rotary shear which is used to cut the ends of the finished bar and also bar length as per marketable lot and is placed after the cooling bed.

20. Electrical Power Supply & Distribution and Instrumentation & Control System

- The Mill Electrical Power Supply & Distribution System mainly includes Transformers, Circuit Breakers, HT Capacitor banks and Control Panels.
- Thyristor Control system for regulating AC motor speed particularly in roughing and intermediate mill stands. In this system the speed of the motor is controlled by stator voltage variation achieved using anti-parallelly-connected thyristor in each phase.
- VVF Drives for regulating AC Motor speed particularly in Finishing Mills is the latest trend, still widely to be adopted by SME-SRRM sector in India. In this the speed of the motor is controlled by varying the supply frequency.
- PC-PLC Instrumentation & Control system for automation of Front & End Cropping Shears, TMT Water Cooling System, Flying Shear etc with valve actuators.

21. Centralized Oil Lubrication System: The Centralised Oil Lubrication System automatically lubricates the gears of the gear box, pinion box etc. The lubricating oil is filtered and cooled and re-circulated in a closed loop.

22. Cooling Water System: Cooling Water System cools the mill stand rolls, fibre bearings etc. The water is filtered, cooled to ambient temperature and re-circulated in closed loop.

1.3 Detailed Technical Specifications for a Standard 15 TPH Capacity Cross Country type RM

Detailed Technical Specifications of Critical Equipments/Parts of a Standard 15 TPH Capacity Cross Country Type Rolling Mill are tabulated at **Table 1.1**.

Table 1.1

DETAILED TECHNICAL SPECIFICATIONS OF CRITICAL EQUIPMENTS/PARTS OF A STANDARD 15 TPH CAPACITY CROSS COUNTRY TYPE ROLLING MILL

S. No.	Critical Equipment/Part installed	Broad Technical Specifications
1.	Roughing Mill	• 22" Mill complete with AC drive motor of 1350 HP, 760 RPM, Squirrel Cage Slip Ring Induction Motor, reduction gear box 1:6 reduction ratio, pinion housing with three output shafts, 3x three high Rolling Mills stands with 22" Dia rolls mounted in fibre bearings in 1st Stand, and antifriction bearings in other stands, 6 T weight Cast Steel Fly wheel installed between Mill Motor & reduction gear box.
2.	Intermediate Mill	• 14" Stand as above complete with 850 HP Squirrel Cage Slip Ring Induction AC Motor, reduction gearbox, pinion housing and five stands in two groups, 1st group of 3 stands and then a speed increaser and 2 more stands after that.
3.	Finishing & Continuous Mill	• 12"/ 10" Stand train with 3 stands in Finishing Mill which are driven by a single AC motor of 450 HP • 2 more stands in the continuous mill driven individually by DC motors of 250 HP capacity each.
4.	Mill Rolls	• Alloy Steel, SG Iron, Chilled CI etc.
5.	Reduction Gear box	• Torsion proof rigid steel fabricated body • Case hardened & ground En 24 pinion spur gear & En 8 helical gears • Splash/Forced lubrication
6.	Pinion Gear Box	• 3HI, High Speed • Torsion proof rigid steel fabricated body • Double helical gears made of EN-19 Quality Steel with roller bearings.

7.	Mill Housing	Mostly Top cap opening type to enable changing of rolls vertically using EOT crane.
8.	Mill Stands	CI or Fabricated MS Steel Stands with screw down mechanism and steel chocks.
9.	Roll Neck Bearings	Spherical Roller Anti friction bearings or Fibre bearings
10.	Gear Couplings	Gear Coupling (Flexible type) made from forged steel installed between Motor & flywheel; Flywheel & Reduction gear box; Reduction gear box & Pinion Gear Box.
11.	Spindles & Couplings	- Each Set of Spindle & Coupling will consist of 1 spindle & 2 Coupling heads. - Spindles made of EN8 Steel & Universal type Couplings.
12.	Repeaters	Steel fabricated oval and square repeaters & pipe nozzles.
13.	Roller Guides	- Cassette/mounted roller guide box: Cast Steel Box with leaf springs, rocker rollers entry guides, lubricating & water cooling system. - Friction guides (open/closed)
14.	Roller Tables	Roller Tables of fabricated MS, with seamless pipes for rollers driven in groups of 8 to 12 rollers at each table, driven by a single motor of 15-20HP for each group through V-Belts or Sprocket and Chains
15.	Tilting/Y-Roller Table	The tilting tables are roller tables as described above. Each tilting table is hinged at one end and lifted by a Pneumatic Cylinder at the other. The tilting table is fitted with a balancing mechanism. In some Units the table is lifted by an overhead motorized winch. The table has to be tilted to take the bar up to enter the pass between the

		middle and top roll. Y- Roller table takes the bar through a sloping platform to the upper pass. The roller table rollers are skewed to take the bar to the Y-Table in front of the sloping platform.
16.	TMT System	This is for martempering the steel bar through controlled cooling of the bar through water sprayed under pressure. The hot bar at about 850⁰C is cooled down to 350⁰C.
17.	Front & End Cropping Shears	Swivel type rotary shear with which is on a movable tray actuated by pneumatic cylinder which brings the housing into the rolling line and out of it after the cutting is over. (Front & End cutting).
18.	Flying Shear	This shear is provided in TMT bar mills, has a pinch roll and cuts material on the fly.
19.	Cut to Length Shear	This is a fixed type rotary shear with 400 mm dia cutting wheel of H11 grade mounted on the machine.
20.	Cooling Bed	A long bed of structural steel (about 30 m length) with air gaps in between to cool the finished bars for further bundling and dispatch. The cooling bed is mostly W-channel type.
21.	Hydraulic/Pneumatic Systems	Pneumatic system for end crop shear, flying shear, Tilting Table operation
22.	Instrumentation & Control System (including PC-PLC)	- Automation of end cropping shears, TMT water system, Flying Shear, Capacitor loading/unloading for power factor improvement. - Thyristor control for regulating speed of AC main mill drive motors. - VVF Drive for Control of Finishing Mill Motor Speed.

23.	Electrical Power Supply & Distribution System HT & LT Capacitor Bank	- The 3 Phase power supply at 33/ 11/ 6.6KV is received from the State Electricity Board (SEB) and distributed within the factory by the Units. The incoming power supply is first metered by the SEB (through CT/PT Metering unit) before being taken into the system's step-down transformers (About 4 Nos.). Each Transformer is normally of about 1000KVA at 440 V load and all Main AC Motors, utilities and lighting loads are sourced from this. The power cables are taken from the transformers into a Main distribution panel, which in turn has leads to all auxiliary drives and their individual panels. Each panel has essentially the following instruments: a) Ammeter/ voltmeter, RPM Meter & PF meter b) KWH meter - HT Capacitor Banks- About 350 KVAR each, installed to control PF of main Motors of Roughing, Intermediate, Finishing Mills. - OCB's & ACB's for every transformer circuit.
24.	Centralized Oil lubrication system	Each of the main mill motor to g/box to pinion housing is served by a centralized oil lubrication System. Each centralized system has its own reservoir of 1500 litres capacity, two pumps (1 running and 1 standby) of 15HP each, Coarse and fine filters for oil(Strainers), Shell & Tube Type heat exchanger to cool the oil, and piping to transport the oil from the CS to the gearbox nozzles and back to the tank by gravity flow.
25.	Cooling Water system	The roll cooling water gets hot during the process. The Hotwell collects the return water flowing by gravity from the roll stands. The water is pumped from the recirculation tank to the stand cooling water headers. The pumps are normally 3 Nos of 40 HP each. Additionally there is a separate cooling water tank for the TMT system since the quality of water is far superior and also the pumping pressure is higher (7.5 – 10Kg/cm²).

26.	Major Auxiliary Motors	a) Pinch Roll Motor (2 x 60 HP) b) Shear Motor (1 x 60HP) c) Tail Breaker Motor (2 x 60 HP) d) Hotwell Motor (3 x 25 HP) e) Conveyor (Roller Table)Motor (4 x 10 HP) f) Cold Shear Motor (1x 30 HP) g) End Cutting M/c Motor (2 x 10 HP) h) Notching M/c Motor (1 x 15 HP) i) EOT Crane Motor (2 x 30HP)
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1.4 GRADES OF INPUT MATERIAL & FORM

- The SRRM units in India mostly process Mild Steel and a few also manufacture products based on Alloy steel.
- Mild Steel is a Low Carbon Steel having carbon in the range of 0.15% to 0.3%.
- Alloy steel contains alloying elements other than Carbon such as Nickel, Chromium, Vanadium, Molybdenum. Low Alloy Steel has alloying elements less than 8% and High Alloy steel more than 8%.
- The Input material is in the form of an Ingot or a Billet.
- Various Sizes of Ingots are used i.e. 4" x 3"; 5" x 4" 6" x 5" etc. and the length is approx 1.37/1.5 m.
- Typical Billet Sizes include: 100 x 100mm; 160 x 160mm etc and the length is approx 1.5 m.
- The prominent grades of Mild Steel rolled include E250, E350 etc.
- The prominent grades of Alloy Steels being rolled include:
 - HD Steel : H11, H13
 - SS : 410, 420
 - Alloys : EN24, EN31, EN8 etc.
 - CD Steel : D2, D3 etc.

1.5 TYPICAL MILL PRODUCT RANGE

- The main products of SRRM include Structural Sections and Reinforcement Products (Rods/Bars).
- The Structural Sections mainly include Angles (100 x 100 mm; 50x 50 mm etc); Channels (100 x 50 mm, 200 x 100 mm etc); Flats (200 x 16 mm; 75 x 12 mm etc), Rounds (100 to 250 mm diameter).
- The Reinforcement Products include small ribbed rounds cold twisted bars in squares and TOR, TMT bars etc typically in 6mm-25mm diameter range.
- Typical Input & Corresponding Product Mix and Sizes being rolled by SME-SRRM Sector in India mainly in Mild Steel are tabulated at **Table 1.2.**

Table 1.2:

Typical Input & Corresponding Product Mix and Sizes being rolled by SME-SRRM Sector in India

12" Mill		16" Mill		20" Mill	
Input	Products	Inputs	Products	Input	Products
<u>Ingots:</u> • 3½" x 4½" • 4" x 5" • 4½" x 5½" • 5" x 6" <u>Length</u> : 1.37m	<u>Angles:</u> 100 x100, 90 x 90, 75x 75, 65 x 65 & 50x 50 mm <u>Channels:</u> 100 x 50 & 75 x 40 mm Flats: 200 x 16 & 75 x 12 mm	a) Ingots: 3½" x 4½" Size & Length 1.5 m b) Billets : 75 x 75 mm, 100 x 100 mm, 150 m x 150mm Size & Length - 1.5 m	a) <u>MS Angles</u> 35 x 35 x 5,6 mm, 40 x 40 x3, 4,5,6 mm 45 x 45 x 3, 4,5,6 mm 50 x 50 x 3, 4,5,6 mm 55 x 55 x 4,5,6 mm 60 x 60 x 4,5,6,8,10 mm 65x 65 x 4,5,6,8,10,12 mm 70 x70 x 4,5,6,8,10,12 mm 75 x 75 x 4,5,6,8,10,12 mm 80 x 80 4,5,6,8,10,12 mm 90 x 90 x 4,5,6,8,10,12 mm 100 x100 x 4,5,6,8,10,12,16 mm 110 x 110 x 5,6,8,10,12, 16 mm b) <u>MS Unequal Angles</u> 45 x 30 x 4,5,6 mm c) <u>MS Channels</u> 75 x 40 mm 100 x 50 mm d) <u>MS Flats</u> 50 x 5,6,8,10 mm 65 x 5, 6, 8,10,12mm 75 x 5,6,8,10, 12 mm 125 x 6, 8,10,12, 16, 20mm 150 x 6,8,10,12,16,20 mm	<u>Slabs:</u> 125 x 150, 140 x 165, 150 x 175, 160 x 160, 200 x 160, 225 x 160, 165 x 190, 175 x 200, 200 x 225 310 x 210 mm <u>Blooms:</u> 250 x 250 300 x 300 & 350 x350 mm <u>Length=</u> 1.5 m	<u>Angles:</u> 110 x 110, 130 x 130, 150 x 150 200 x 200 & 250 x 250 mm <u>Channels:</u> 200 x 100, 250 x 125, 300 x 140, 125 x 65, 150 x 75, 200 x 75, 300 x90 & 400 x 100 mm <u>Rounds</u> 100 mm to 250 mm diameter

CHAPTER 2

STANDARD OPERATING PRACTICES IN ROLLING MILL

2.1 RAW MATERIAL SECTION

2.1.1 Receiving, Testing & Stacking of Ingots/ Billets

- Ingots/Billets received at the Rolling Mills come with a Delivery document which gives details of the Party, Total weight, Number of pieces, Grade, Chemical analysis, Size, relevant ISS number. The Ingots/Billets are then counted & weighed at the Unit's weigh scale and they are handed over to the raw materials -in-charge (RMIC). 100% visual checking for surface defects is done.
- The quality of the Ingots/Billets is checked by the QA group who determine the suitability of the Ingot/billets for further rolling without problems. Two samples are taken from each lot received at the Rolling Mill on random basis, small pieces cut and subjected to Chemical Analysis using spectrometer and percentages of C, Mn, Si, S, P are ascertained to ensure that they are in line with the Chemical Analysis Report which came with the Ingot/Billet lot.
- The Ingots/Billets are then stacked in the Raw Material Storage Yard lot wise with each Ingot/Billet painted with appropriate Colour Code and information i.e. Heat No, Size, Material Grade marked.
- Each stack should ideally be 1.8m high, about 10 billet/ingot thickness width (approx 1.5 m) and 1 billet length long (approx 1.5m).
- For easy movement of Manpower & equipments (crane/fork lift) between stack rows there should be 0.7m spacing widthwise & 3.5 m lengthwise.
- A Stack card is prepared by the RMIC which contains the following details for each lot/stack:
 - i) Batch/Source/Lot no.
 - ii) Date of receiving
 - iii) Number of Ingots/Billets received, Size & Grade
 - iv) Total Tonnage

- v) Stack Location
- vi) Stack no.
- vii) Number of Layers per stack

2.1.2 Ingot Preparation

- If found necessary, the QA Staff gets the Ingots trimmed at the back end to get rid of excessive piping areas. Sometimes if the ingot has excessive piping, then the ingots are not sent for rolling but returned to the supplier as such ingots could cause immense problems during rolling through splitting etc.
- The size of the ingots to be rolled is decided based on the finished sizes to be rolled out of the ingots and could be 3"x4", 3 ½" x 4 ½", 4 ½"x 5 ½". The lengths of the ingots can be specified to the supplier wherever possible to increase the yield. The number of pieces rolled out of a single ingot should be as far as possible an exact multiple of the finished product length required, to avoid wastage due to short lengths.
- While determining the size of the ingot, due consideration should be given to the end cuts while rolling, burning loss etc and the final yield expected per ingot. Where small sections are rolled, many mills cut the ingots in two pieces to accommodate the length achieved during rolling at each stand and the available space at the stand to accommodate the longest length arising. Smaller end products (<10mm) result in higher temperature drops during rolling.
- All fins, scrappy ends etc are trimmed to prevent falling off of the same during rolling and getting stuck in the guides.

2.1.3 Billet Preparation

- The size of the billets is decided based on the finished sizes to be rolled out of the billets and could be 110x110mm, 100x100mm or 130x130mm or 150 x 150mm. Finished weight means the weight per piece of finished length.
- The weight of the billet selected should be as close as possible to a multiple of this finished weight, so that the wastage due to short lengths is practically reduced to zero. The maximum length of the billet is the width of the Reheating Furnace. The cross sectional size of the billet x length of

the billet is limited by the maximum length that can be rolled in the roughing stand. The normal layout of the mill limits this length as the bar being rolled in the roughing mill could interfere with the intermediate mill train on the outgoing side or on the ingoing side the Reheating Furnace discharge roller table. Once the bar enters the repeater the length does not matter.

- The Billets of desired size are cut from Blooms of larger dimensions (cross Sectional area & length) and in order to increase the yield of the billet, this cutting should be done using a band saw, where the wastage is limited to the width of the saw blade i.e. around 1.5 mm, compared with the weight of billet lost by gas cutting of minimum width of 5 mm per cut.

2.2 Product Planning and Scheduling

- Production Planning is dependent on several factors like:
 1. Orders in hand & present stock
 2. Production per day
 3. Sizewise breakup of Orders
 4. Present rolling schedule
 5. Residual life of the roll pass on current rolling size
 6. Size to be changed to so that least number of rolls/roll passes require to be changed
 7. Raw material availability
 8. Cash & Carry customers
 9. Planned shutdowns downstream
 10. List and gradation of Customers for fixing priorities:
 - a. Gradation depends on
 - i. Past punctuality in payment-Grade A
 - ii. Sizeable order size-Grade A
 - iii. Future potential with respect to advantage to the Company-Grade B
 - iv. Reach to other customers-Grade C
 - v. Margins influence decision making -sizes with higher margins are prioritized for earlier rolling -Grade A
 - vi. Unit selling price above market price -Grade A
 - vii. Unit selling price at par with market price -Grade B
 - viii. Fault finding tendency to bargain for getting reduction in the Selling Price-Grade C

b. List of Customers whose Orders have been booked and advance received -Priority A

- All the above factors are carefully weighed and after careful consideration the rolling sequence is arrived at and the schedule is transmitted to the Shop Floor and Marketing Department. Planning once made is not to be disturbed for at least a week if not one month.
- The Format for the Weekly Production Planning Sheet which should be prepared by the Planning Department for the GM of the Rolling Mill is provided at **Table 2.1.**

Table 2.1
Format of Weekly Product Planning Sheet

Planned Date of Rolling.	Size	Weight to be rolled	Grade & Heat No: Stacked at:	Destination: Customer name/ Finished Goods Store	Special Processing Instructions & Stacking instructions

- For efficient tracking of each Lot/Batch right from Raw Material Stage till dispatch a Heat Card needs to be maintained where Heat wise all necessary details need to be recorded by all Concerned Departments and in the end of the processing cycle this Card is returned back to the Production Planning Department/Management for record & analysis purpose. The recommended Heat Card Format is provided at **Table 2.2.**

Table 2.2
Heat Card Format -Ingot/Billet Processing

Heat Card Sl No:	Grade: Analysis:			Stacked at: (location)	
Heat No: Date & Time of Handing Over to fireman:	C Mn Si S P	As per supplier	As per own Lab	Size: Heating Cycle code:	Qty: Nos: Weight: Weigh slip No:
Received by: (In-Charge RM):	Recd Test Certificate: Qty recd: Nos kgs				

Disposition of Raw Materials:		Ingot /billet preparation: Cut to size:	
Charged into RHF:			
Date:	Qty: Nos Kgs	Balance in hand: Location:	
ROLLING SIZE: Date: a) mm b) mm c) mm	Input Weight: a) kgs b) kgs c) kgs	Finished Wt: a) kgs b) kgs c) kgs	Yield: a) % b) % c) %
Finished Goods Stacking: Colour Code:	Handed over to dispatch Section:	Date:	Received by:
Size: mm Wt: kgs Qty: Nos	Stacked at:	Dispatched to: Date of Dispatch: Wt dispatched: Balance in Hand:	Tag details:
Accounts Dept:			
Verified by:			

2.3 SETTING OF ROLLING MILL BEFORE OPERATION

2.3.1 Roll Turning, Roll Pass Schedule, Roll Pass Design

- Roll Pass Schedule is the number of Drafting steps & size and shape of Rolling Passes to reduce the Input Billet/Ingot into finished product/section of desired shape, Size at the end of the Finishing Mill/Continuous Mill.
- The roll pass design for any product depends on the following:
 - Starting size & material grade
 - Mill layout
 - Mill stand sizes
 - Mill motor power
 - Production requirement
 - Product size& shape

- In order to ensure smooth rolling for a given roll diameter, the reduction in a pass is decided in such a way as to keep the bite angle within acceptable limits i.e. $18-22^{\circ}$.
- Typically, a pass design calculation has three parts :
 - Pass design and groove details
 - Pass schedules
 - Power calculation
- Pass Design and Groove Details: This calculation gives the following parameters for each pass:
 - Roll groove dimensions
 - Roll gap
 - Filled width in pass
 - Filled area
 - Area reduction
 - Bite angle
- Pass Schedules: Pass schedule consists of the following for each pass:
 - Bar length
 - Rolling speed
 - Rolling time
 - Idle time
 - Loop or tension value between stands
- Power Calculation :Power Calculation works out for each pass:
 - Bar Temperature
 - Rolling load
 - Rolling torque
 - Rolling power
- Computerized Mathematical Models & Programs are available that can efficiently be utilized by the Roll Pass Designers to optimize the Roll Pass Design for given Set of Inputs & Finished Products. The Mathematical Models for the Rolling Mill incorporate all the above mentioned Roll Pass Design parameters.
- Based on the Roll Pass Design, each Roll in the Stands of Roughing, Intermediate and Finishing Mill is grooved.

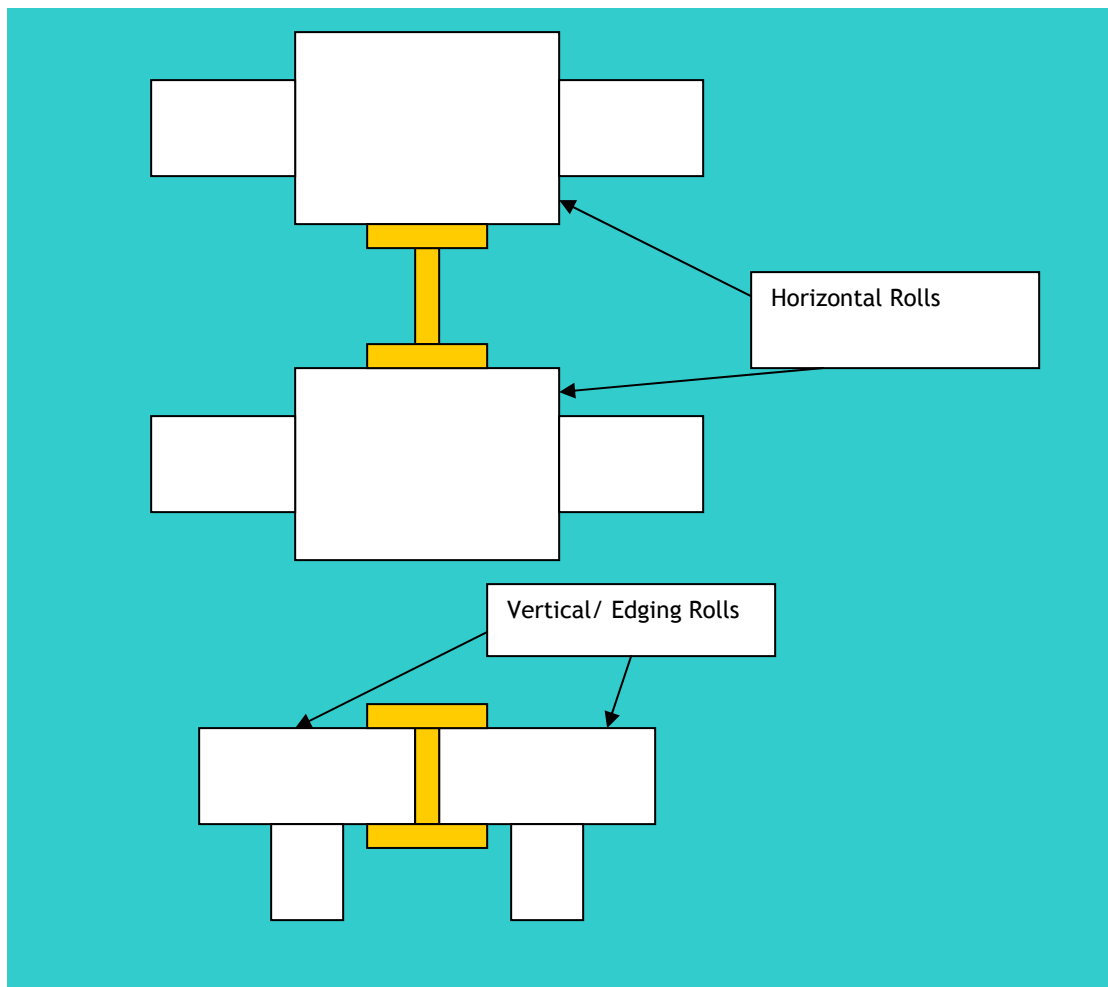
- The various shapes of grooves are:
 - For Breakdown passes in Roughing Mill: Sequence Box Grooves (per pass reduction ratio of 0.35-0.40); Sequence-Square-Diamond-Square type (Per pass reduction ratio of 0.15-0.20).
 - After Breakdown passes the sequence can be:
 - i) Diamond -Square -Diamond (per pass reduction ratio of 0.15-0.22)
 - ii) Square-Oval -Square (per pass reduction ratio of 0.20-0.30)
 - iii) Round-Oval-Round (per pass reduction ratio of 0.12-0.25)
- Roll grooves machining instructions should indicate the radius of corners and relief at the sides to prevent sharp edges.
- All templates for the different roll passes are numbered by punches for identification.
- Templates to be used by the roll turner for checking the passes under machining are to be indicated to check the roll pass being machined and should be available with the Roll Turner.
- During re-machining of the roll pass, the roll grooves should be machined to the extent of removing all traces of firecracks from the groove. The pass having the maximum depth of firecrack should be machined first and the collar diameter is then known and other grooves machined down using this collar diameter as the reference.
- An accurate way of turning the rolls is by using copy turning attachment which has a stylus with the current Manual Lathes after conversion that traces the profile of the template and guides the tool holder accordingly.
- The tool tips are either sintered carbide tips or HSS. Whenever the tool gets worn out it is ground in a profile grinding wheel.
- The finishing pass is generally ground finished to give a smooth surface finish.
- For machining rolls used for section rolling, lathes that have the provision for mounting matching roll together with the roll being machined, together on the lathe bed one above the other should be used for accurate machining of the matching rolls.

2.3.2 Draft Adjustment

- The amount of reduction in area by passing through a roll pass and expressed in percentage of the input size to that pass is known as draft.
- Draft depends on the motor power, roll quality and diameter and stiffness of the stand.
- The stiffness of a stand is the resistance to deformation of the stand under rolling pressure. Mill Spring is the Stand Deformation that takes place under rolling loads. The angle of bite in a pass should be such that the bar enters the pass without hesitation. The angle of Bite is tried to be maintained less than 22° . When determining the depth of roll pass, the amount of spring in the stand is taken into consideration.
- When the input size is large compared to the roll diameter by increasing the angle of bite, the bar will find it difficult to enter the pass. As a thumb rule the maximum size of the input material should not be greater than 0.35 roll diameter in mm for a good bite angle at entry.
- To overcome the biting problem the roll pass can be ragged either by a knurling tool during roll turning or by welding high points on the roll pass which help to drag the bar into the pass through friction, without leaving large indentations in the bar. However to avoid knurling or welding in Roll passes, it is recommended to maintain a proper Bite angle by maintaining proper Roll Diameter.
- If the indentations on the rolled bar caused by the welded beads are deep then there is every possibility of lap or folding over of the indentation occurring later in the rolling. This is a surface defect which is carried into the finished product and could lead to rejection, especially when rolling alloy steels.
- While high draft is necessary for rolling alloy steels to break down the grain structure it not so important in rolling mild steel, except that it determines the number of passes required to arrive at the final size. While rolling high alloy steels it is necessary to break the grains in the initial passes, and for this the diamond square diamond passes are ideally suited. The diamond passes can cause roll breakage because of excessive depth of the pass and the ratio of depth of pass to roll diameter should not be greater than 1:4.

- Draft should be such that the Mill motor can take the load, the roll strength (stiffness or Length: Diameter of barrel ratio) and should be such that the roll does not break or deflect beyond 0.001” per inch of roll barrel length. SG Iron & Cast Iron rolls deflect far less than Steel Rolls and hence they maintain the section better.
- The depth of the roll groove should not become a weak point inducing rapid fatigue strength life leading to early roll breakage. The inner diameter of the Roll should not be less than the Core diameter of the Roll.
- In motorized screwdown operated stands the depth of pass for rolling heavier sections can be made very small and gradually the top roll is brought closer to the bottom Roll in subsequent passes.
- Special care should be taken while designing roll passes for rolling sections like angles, channels or I-Beams, where there would be sharp corners at considerable depth in the rolls. The reduction ratio is greatly reduced in such passes. Now edging roll assemblies with grooved rolls are available for mounting on restbars, which can take care of light edging requirements while rolling light sections.
- For section rolling it is advisable to have an edging roll or otherwise known as vertical roll stand after two horizontal roll roughing stands and again one more after two more horizontal roll stands (**Ref. Figure 2.1**). This is to give the proper profile to the I-Beam and C -Channel sections to maintain parallelity of the flanges and perpendicularity of the web. This also saves on Rolling Mills cost where large diameter rolls are required when edging rolls are not present, because the edging function has to be performed within the horizontal rolls themselves leading to wastage of roll diameter and extra power of the motors to drive such large rolls.

Figure 2.1: Illustration- Horizontal and Vertical/Edging Rolls Assemblies



2.3.3 Roll Changing

- For changing the rolls, the Rolling mill motor is stopped and a shutdown is taken on the Mill drive and only the inching operation is made available to the pulpit operator.
- The top screwdowns are loosened, the balancing springs are de-latched and the top caps are opened out by driving out the cotters or opening out the cap holding bolts.
- All spindles on the drive and free ends of the roll assemblies are supported, at their centres, on a stand fabricated for the purpose of roll changing.
- The spindle couplings are disconnected from the roll journals and the spindle end covers are drawn back and tied to the spindles.

- The guides and guards are loosened and taken back on the rest bars to give a clear path for the rolls to be lifted out.
- The roll and bearing housings are lifted out by crane slings and deposited at the roll shop. The slings are leather encased to protect the roll passes from getting damaged. Some Mills have their bearing chocks with two holes with threads drilled and tapped for screwing on lifting shackles. A special lifting tackle with four equal leather encased chain slings with shackle rings at one end and hooks at the other are suspended from this tackle, which itself has a single shackle ring which is taken up by the EOT Crane Hook on the main hoist. Where such tackles are not provided, two leather/ thick nylon sheathed steel slings with eyes at both ends are put around the roll barrels and the eyes are put on to the EOT Crane Main hoist hook.
- The roll assembly complete with the bearing housings is lifted out vertically from the stand housing and taken straight to the Roll Shop and deposited there. The other two roll and bearing housing sets are taken out in similar fashion.
- The new bottom roll and bearing housing assembly are then lifted from the roll shop and lowered into the mill housing and placed on top of the bottom breaker blocks. The middle and top roll assemblies are fitted into the mill housings similarly.
- The top breaker blocks are then placed on the top bearing chocks and the top caps are fitted back, the balancing springs are latched on and the cotters tightened. Breaker blocks are round CI 50mm thick discs that crush and collapse on excessive load on the rolls and greatly prevent roll/ bearing/ spindle breakage.
- The screwdowns are operated to touch the top screws onto the breaker blocks.
- The guides, guards & strippers are fitted and the cooling water pipes are now fitted.
- The major steps in roll changing are illustrated at **Figure 2.2** and the critical parts of the Roll Lifting System are depicted at **Figure 2.3**.

FIGURE: 2.2 - Steps in Roll Changing

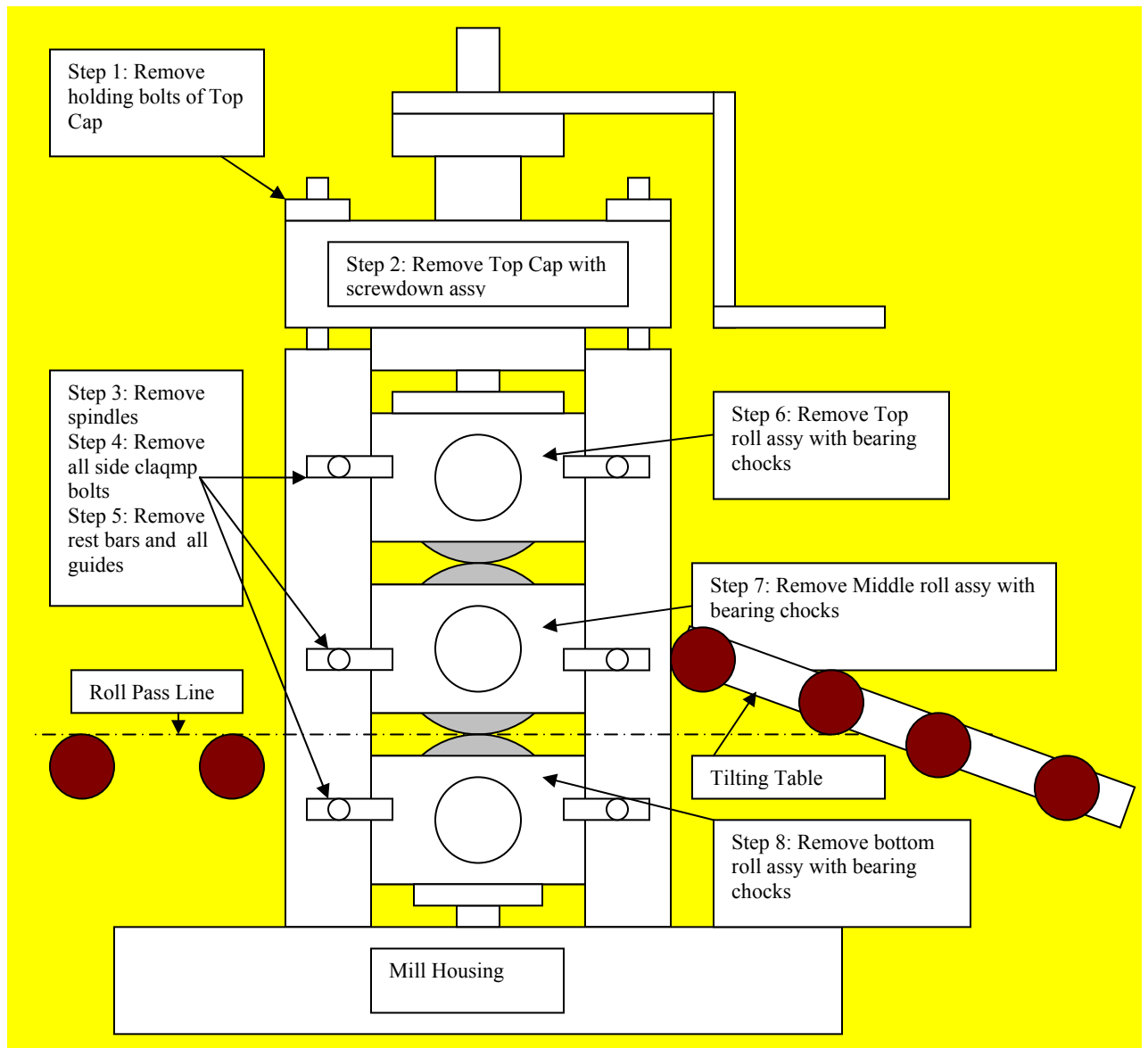
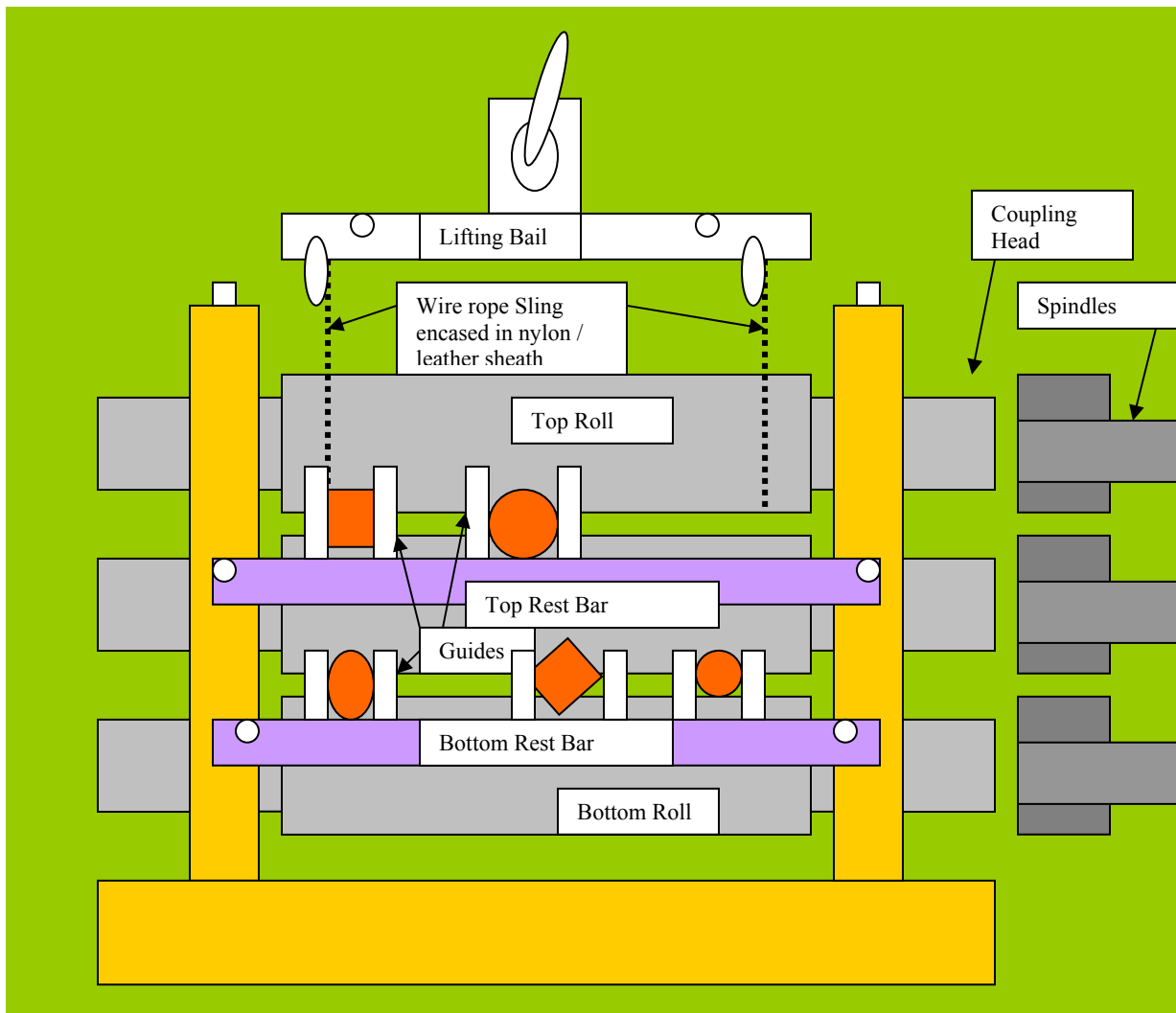


Figure 2.3: Illustration –Critical Parts of Roll Lifting System

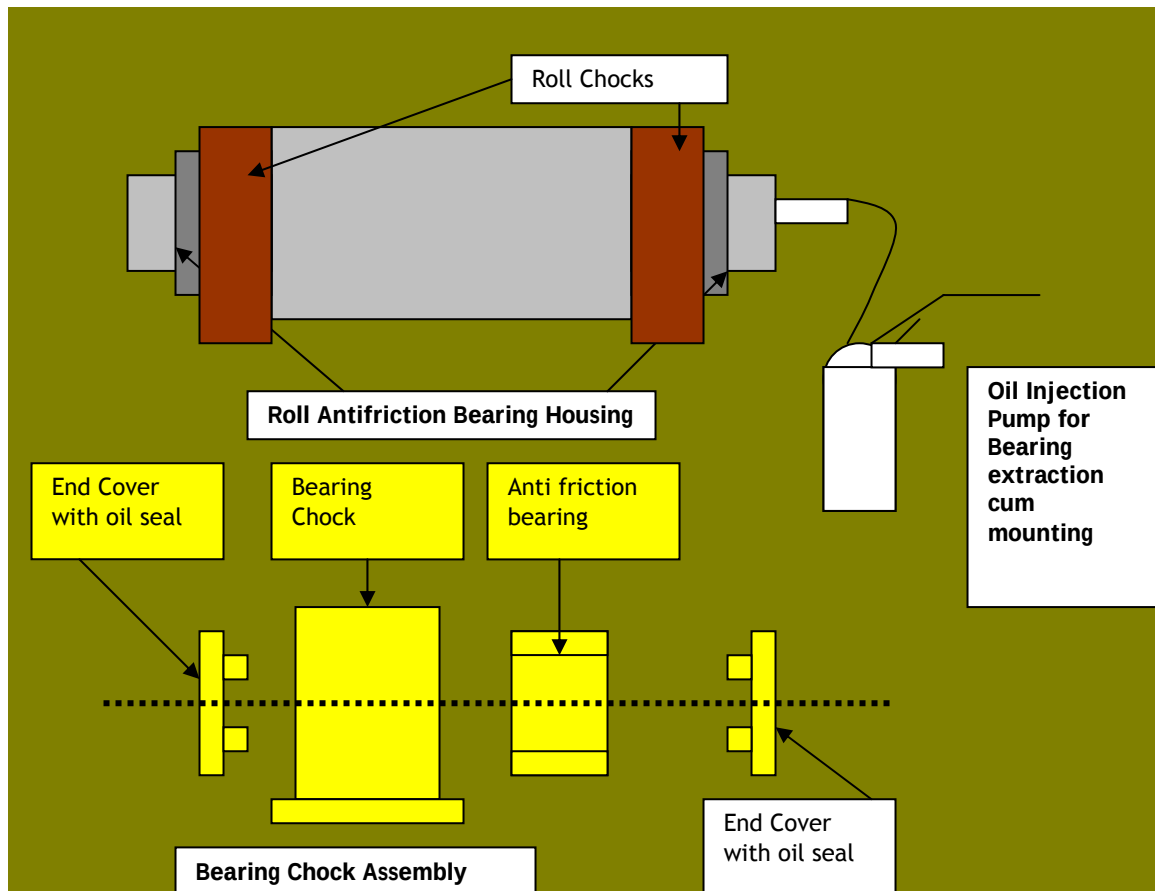


- **Dismantling the Bearings from the Roll necks:** If the roll neck bearings are fibre then the dismantling is easy and the bearing halves are taken out of the bearing chocks and new bearings inserted.

If the roll neck bearings are antifriction or taper roller bearings, then the bearings are dismantled by applying hydraulic pressure between the roll neck and bearing inner race, using SKF oil injection system. The bearing outer race is a sliding fit in the bearing chock and the chock can be slid out first. The oil injection pipe is fitted onto the roll neck. There is a hole in the roll neck leading to the centre of the inner race of the bearing. Oil injection pump is then pumped and the high pressure developed (350kg/cm²) expands the inner race sufficiently for the puller cum

hydraulic jack to extract the bearing out of the roll neck. The Schematic Diagram for dismounting of Roll Antifriction Bearings is provided at **Figure 2.4.**

Figure 2.4: Schematic Diagram for Dismounting of Roll Antifriction Bearings



- Mounting of the Bearings: The mounting of the bearing is done in reverse order on to the roll neck of the roll made ready for the next campaign.

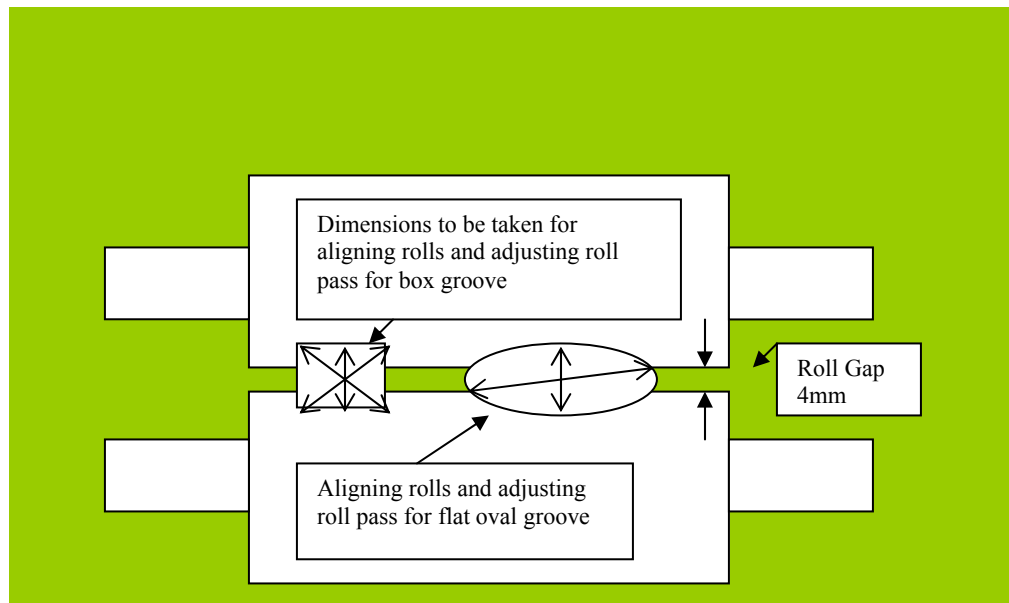
2.3.4 Roll Setting

- After the Roll assemblies with chocks are placed in the Mill stand housing they are set/prepared for rolling.
- The side clamps on the mill housing for the middle roll are now tightened to prevent lateral movement of the rolls during rolling.
- The level of the middle roll is taken as the reference level. The bottom and top rolls are adjusted according to the requirement of the roll pass. The

roller table feeding the mill is adjusted according to the level of the bottom roll. The leveling is done by adding/ removing shims from the lugs on which the bottom roll chocks rest or by adjusting the screwdown mechanism of the top rolls.

- A straight edge is placed on the centre of the roll pass which would be used, and the other end is kept on top of the roll pass line marker, which is the top of the roller table first roll. The height of the rest bar is adjusted so that all the bottoms of the guide boxes are 2mm below the roll pass line or slightly below that.
- The top and bottom rolls are now squared with respect to the roll pass on the middle roll, using an inside caliper and vernier calipers to measure the gap between the roll collars at either end of the roll, as well as the diagonals of the roll pass.
- The Dimensions to be taken for aligning rolls and adjusting roll pass for Box groove & flat oval groove are provided at **Figure 2.5**.

Figure 2.5: Illustration -Dimensions to be taken for Aligning Rolls and adjusting Roll Pass for Box Groove & flat Oval Groove



- The normal roll gap at the collars, when the rolls are newly changed is 4 mm and is measured by a machined steel gauge flat of 4mm thick welded on at the end of a 6mm rod for ease of holding. The pass dimensions are machined assuming that this would be the roll gap. As the roll pass wears

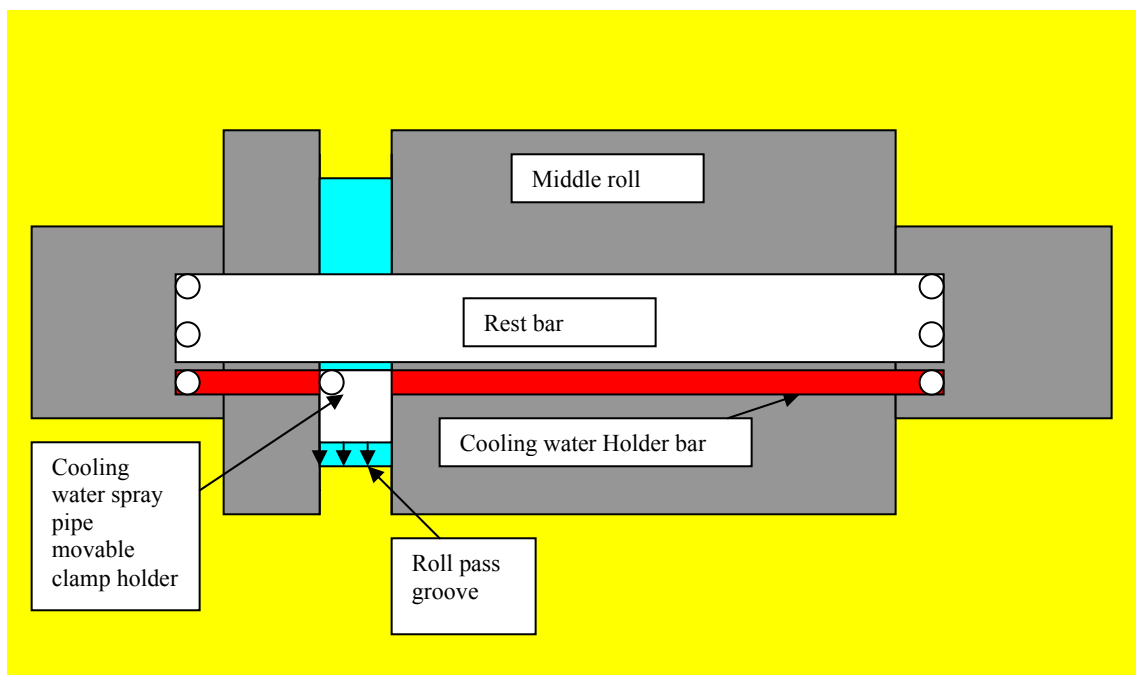
out the roll gap is decreased 1mm at a time until the roll collars touch between middle and top/ bottom.

- The roll gap equalization at either end of the roll barrel ensures that the rolls are exactly parallel and the middle roll being level, the top and bottom rolls automatically become level.
- The alignment of the roll pass between the bottom and middle rolls are firstly checked by holding a lamp at the exit of the pass. A person standing in a trench in front of the Mill (so that he need not bend on all fours), now sights the light and asks the operator to adjust the screws of the lateral window clamps on either side of the housing so that the pass is perfectly aligned and in one line.
- If the pass is a vertical or horizontal oval, then the roll squaring is done as above. If the pass is square or diamond then the diagonals of the pass is measured and equalized as above using the window clamps.
- The guides and guards are set by using the lamp on the opposite side of the person looking into the pass. When in line, the holding bolts of the guides with the rest bars are tightened fully. The same is repeated for the bottom and middle roll settings. The sample bar should pass between the side plates of the guide freely with 1mm gap on either side. This will not allow any scoring marks to come on the bar while passing through.
- The entry roller guide box (RGB) has two flaps with rollers mounted at each end. The flaps are on guide ways separated by a lug wide enough to allow the bar to be rolled through the RGB. The position of the flaps and their angle can be adjusted and locked by lock screws from the top and side of the box. The gap between the rollers of the RGB is so adjusted that when the sample of exact size, which was rolled out from the previous pass, is passed through the RGB it should contact both the rollers and rotate them but at the same time it should be free enough to allow the sample to pass through without extra pressure.
- If repeaters are there then the exit pipe to the repeater from the previous roll pass is so adjusted that the horizontal oval bar or square enters the repeater at a skewed angle and at a height which makes the bar hit the tangent of the repeater arc and continues tilting as it circles around the

repeater and enters the roller guide box vertically. Some repeaters, known as escape type repeaters are pneumatically operated to open as soon as the bar enters the succeeding roll pass, so that the jerk does not transmit to the repeater especially when heavier sections are repeated.

- The cooling water hoses are now connected to the RGB through rubber hoses.
- The Roll cooling water pipes are adjusted by sliding the clamp holding the water header on the underside of the rest bar (for middle roll) so that the water jets are pointed only toward the roll pass. No water is allowed to fall on the barrel. The water is then closed by adjusting the valve on the header to the stand rolls. The Schematic Diagram for Roll Cooling Water Pipes Adjustment is provided at **Figure 2.6**.

Figure 2.6: Schematic Diagram for Roll Cooling Water Pipes Adjustment



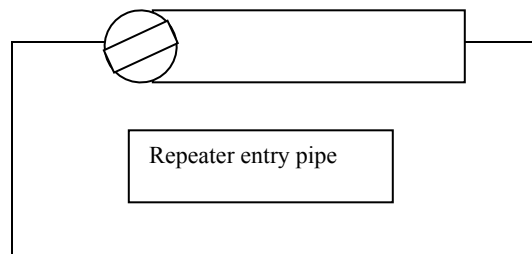
- The spindles are then fitted back and all cover bolts fully tightened.
- After cautioning all the workers and staff on the shop floor by blowing a siren three times, the mill motor is switched ON and the mill rotation is inched forward.

- If there are any obstacles or abnormal sounds then the motor is immediately switched OFF and the cause investigated thoroughly. The most common reason for abnormal sound is that of rubbing of the Guides & Guards with the Rolls. If the Roll Setter has been careless it could be some fallen piece between the Roll and the Rest Bar/Guide.
- The obstruction is cleared and a note is made to include the elimination of all such mishaps in the future and the motor inching restarted.
- If everything is normal then the Mill is now declared to be ready for rolling, by blowing the siren twice.

2.3.5 Pass Burning

- After Roll Setting and starting of the Motor, Pass burning of Mill is done with hot and soaked front end chamfered trial pieces for the following purposes:
 - The pass surfaces of individual passes are smooth after roll turning is done and biting becomes a problem because the bar slips at the entry even though the angle of bite is correct. For better and trouble free rolling the pass shall be roughened by passing hot samples through the new pass to ensure better biting.
 - Collars of working passes shall be matched by adjusting the lateral movement of the rolls through tightening or loosening the side clamp bolts of the roll to fix them properly to avoid defects during rolling.
 - Exact amount of mill springs (deformation of the Mill housing under load) shall be known in Stands for factoring in this for the fine adjustment of roll gaps.
 - The amount of spread due to mill spring shall be known in a particular stand.
- The Main Mill drive motor is started.
- The samples to be used are usually pieces of misrolls that have come out of the previous roll pass. These are cut to suitable sizes with front end chamfered and kept in a shelf next to the Mill Stand and catalogued by painting the size on the sample.

- If no such sample is available the sample is prepared by rolling a piece through the previous passes.
- The sample is heated in the soaking zone of the Reheating Furnace and when the temperature is around 1050⁰C the sample is quickly brought to the roll pass manually by a person holding it with tongs.
- The sample is passed through the roll pass and the piece is allowed to cool.
- It is ensured that during Pass Burning there is no cooling water supply to the Rolls & Guides.
- Measurements are taken of the rolled sample bar and compared with the design dimensions.
- Any error on dimension and shape is corrected by moving the bottom/ top roll up/down/sideways for fine adjustment of Roll gap.
- If the bar is tight or too loose in the guides then the guide flap opening is adjusted accordingly.
- Another sample is then tried in the same pass. If the dimensions and shape are OK then the same is repeated on all the passes.
- The height of the repeater and alignment are adjusted to receive and deliver the bar sample smoothly.
- The exit pipe from the roll pass leading into the repeater is skewed to lead the bar at an angle into the repeater so that the bar turns by 90deg by the time it exits the repeater and enters into the entry guide of the next pass on the next stand vertically.



- Typical instructions to be followed for adjustment of RM during Pass Burning for rolling of Rounds is provided below:
 - If diameter is too small: open the gap between rolls.

- If diameter is too big: Close the gap between rolls by screwing down the screw down.
 - If there are side fins: Reduce the thickness of the leading oval.
 - If there is flatness at the sides: (underfilling of pass): Increase the thickness of the leading oval.
 - If there is seam or fin on one side and underfill on the other side: The entry guide is eccentric. Adjust the entry guide by moving it to the centre by using a light at the other side and checking the guide profile.
 - If the hemispheres are displaced to one side: Centre the top and bottom rolls by adjusting the window clamp bolts.
- One Billet shall be taken and its behaviour shall be observed during rolling. Any deviation in stock dimension if observed at the intermediate or finishing stages, the Rolling Mill setting should be rectified. This process shall be continued till the final section is achieved.

- **Rolling Sections:** The above procedure holds good for section rolling also.

The checking of sections for the various passes in section rolling is done by using templates for top and bottom sides of the rolled piece. Common problem faced in the rolling of sections is that one side of the angle/ channel becomes too heavy. This is because of improper centering of the bar in the forming passes maybe due to bad positioning of the entry guides and/or due to improper soaking of the ingot/ billet. It is very important to keep the scrap diameter of the rolls in mind while cutting the grooves for sections. The core diameter of the roll is softer than the surface which attains hardening due to indefinite /definite chilling procedure adopted during the manufacture. Hence while grooving the rolls the core diameter should never be reached.

- The Mill shall then be considered set and continuous rolling shall be started.

2.3.6 General Instructions before Rotating Rolls/Check List for Rolling Supervisor

- Before rotating the rolls after setting, the following should be ensured:
- o Check the rolling Programme for the day from the Production Planning Department.
 - o Check position of ingots/ billets inside the RHF for grade and quantity and check if it matches the rolling programme received.
 - o Check for the availability of ingots/ billets for charging during the shift

- o Check the temperature of the furnace for readiness to roll.
- o Set the Mill for rolling. Check all the rolls in all the stands, which have been prepared by the night shift workforce who have the duty of keeping the Mill ready for Rolling the next morning.
- o Check the positioning of all the guides, roller guides, guards, stripper guides and repeaters and tighten all bolts.
- o Check all the roll clamp bolts after aligning the rolls
- o Check all the roller tables for free rotation and by power
- o Check the settings of the cooling water hoses, temperature and pressure at each Stand.
- o Check that proper Flow, Pressure & Temperature is maintained for the Oil Lubrication system, after checking level of oil in the Tank.
- o Check the functioning of all the Interlocks.
- o Check with the Electrical department that all the drives are functional
- o Check with the Mechanical department that all the equipment are in good operating condition.
- o Check for the sufficiency of manpower at each operating station.
- o Check that the Mill Floor is kept clean of any cobbles from previous shifts.
- o Check the operation and settings of the TMT box. Check that all the instrumentation is connected and ready for operation.
- o Check the EOT Crane on all motions.
- o Check the spare rolls position at the roll turning shop and ensure that at least one set is available for each stand for replacement in case of breakage/ wear out.
- o Check the samples kept inside the furnace for setting the roll passes
- o Ask the Electrical Department to start the equipment one by one. When all the equipment are running smoothly, the first sample is taken after shutting off the water to the roll pass cooling. Check the rolled sample bar for dimensions as per schedule.
- o Pass samples through each and every pass and check for dimensional correctness of the exiting bars, both in Hot and then cold condition.
- o Open the water supply for roll cooling.
- o Take the first ingot/ billet from the furnace. If it reaches the cooling bed take all the dimensions for two samples cut from this bar -one at the front end and the other at the rear end. Take the weight of the samples and adjust if necessary.
- o The mill is now ready for rolling.

2.4 STANDARD OPERATING PROCEDURES FOR ROLLING MILL

2.4.1 Rolling Supervision & Section Control

The important Aspects to be kept in mind during rolling supervision are:

- A constant watch is maintained to see that the billets fed into the roughing mill do not have heavy piping which could pose subsequent problems in the mill. Piping causes the collapse of the bar at the spots wherever it exists as there is no steel at these spots except on the shell.
- The uniformity of heating of the billet throughout its length and cross-section is also watched and surface temperature measured, using an infrared temperature measurement meter, on sample basis to ensure correct rolling temperatures and immediate corrective action is taken by the Reheating Furnace operator, who will adjust the burners on the endwall. The temperature of the bar during the rolling process is measured especially at entry of the ingot/billet at Std#1, entry at finishing stand/TMT Box and at 10m distance from TMT Box on the cooling bed.
- The length of end cuts are also adjusted to the minimum so that the purpose of cutting the ends to prevent splits is achieved as well as the purpose of increasing the yield, which means an extra effort at arriving at the correct minimum length that needs to be cut.
- The ends are cut at both ends of each bar at exit from the Std#1 to prevent splits in the bar during subsequent rolling, especially when repeaters and roller guide boxes are used. For end cuts an end cut crop shear is installed at either end of Std#1. Two automatic swiveling type rotary blade end cropping shear can also be installed at entry to the intermediate mill or Std#4, one shear cuts the front end while the other cuts the rear end while the bar is in motion.
- Whenever a misroll occurs, it occurs mainly due to the temporary negligence of the supervisor. A good supervisor removes the root cause of a misroll before starting the rolling process by proper mill setting as explained above and enhance Mill available hours.
- The Mill supervisor also watches for any minute changes in the behaviour of the bar being rolled, whether the size coming out of any pass is larger

than normal, whether the bar is having fins or overfilling of pass, whether the bar is finding it difficult to enter the next pass, whether any guide or guard has worked loose, whether the end cuts are longer than necessary, whether the end cut is clean and there is no end piece not getting cut fully and gets carried over a short distance towards the mill, whether the front and back end difference in unit weight is as minimal as possible.

- Section Control commences with the roll setting at the commencement of the shift. The supervisor should have a checklist with him on all the parameters involved in roll setting and the size of the bar to be achieved after each pass as per the Roll Pass Schedule.
- Samples must be passed through all the passes and the section of the sample coming out of that pass should be as per that required by the next pass. The entry of the sample into the pass should be smooth, the guide rolls should be set at the exact width that allowed smooth entry without any chance of the bar tilting or hitting the roll pass shoulders.
- The profile dimensions of the bar coming out of the Roughing, Intermediate & finishing Mills pass should be measured on a random basis after cooling the bar in water and comparing it with that required.
- During rolling the supervisor should keep a lookout to see that there are no fins on the bar caused by overfilling of the pass or flattened sides of the bar due to underfilling. Each and every pass is important and should deliver the exact size of each bar. One should not wait for subsequent passes to correct the wrongs of the previous pass. This would invariably lead to a misroll.
- The unit weights of the front and backends (1 ft long) should be taken at least once every two hours of rolling. If the difference is more than 1% then corrective action should be taken like:
 - Uniform heating in the Reheating Furnace and proper soaking of ingots/billets.
 - Correct temperature of the billet at exit from the Furnace.
 - Correct end cuts at the crop shears.
 - Make sure that there is no jumping of the top roll when the bar enters the finishing passes by adjusting the screwdow and ensuring there is no backlash in the threads.

- The Dimensions of the finished products (Side of Square bar, diameter of round, Side of Angle etc) and also angles need to be monitored using Micrometer, Vernier Calliper, radius gauge etc by cutting about 1ft long piece from bar centre, once every hour of rolling , after cooling in water.

2.4.2 Emergency Stopping Of Mill

- In case there is a misroll which is caused by the collaring around the roll, or if there is any other situation which calls for an immediate stoppage of the Mill, then the pulpit operator presses the emergency button to bring the rolls to an immediate stop. This would set off the siren to warn everybody in the Mill and at the Reheating Furnace about an emergency in the Mill.
- The furnace operator stops feeding any more billets and if PLC or PID Controller exists then he takes the necessary action by pressing the controls equivalent to shutting down of the furnace temporarily.
- The roll Setting Staff on the shop floor rush to the stand that caused the problem and work to restore normalcy. The cause of the problem is thoroughly investigated by the supervisor and corrective measures taken and if necessary resets all the previous mill rolls before taking the next bar.

2.4.3 Action taken in the event of Cobble

- A cobble or misroll is a bar not passing through the finishing stand with the correct size and weight, once it has been discharged from the furnace. In the event of a cobble or misroll, the mill is stopped only when considered absolutely necessary. Otherwise the corrective action like clearing the mill of the cobble and checking of guides and bolts etc. are done while the rolls continue to rotate.
- In case the bar is badly stuck then the bar is gas cut into pieces and then removed. At this time the motor is stopped before continuing to remove the cobble.
- The guides are checked for looseness. The guides are suitably tightened to allow the bar exactly to enter the pass. The leading end of the cobbled bar is examined to see whether there was a split which prevented the bar from entering the guide/ pass. If the phenomenon of split has been occurring for few of the preceding bars, the length of end crop should be increased slightly and the next bar front end is examined to see whether the split has

disappeared. The process is continued until there is no sign of a split. Caution must be exercised to see that longer than absolutely required crop end is not cut for that would increase the losses and decrease the yield considerably.

- To prevent collaring of the bar the root cause of avoiding rolling split bar ends by cutting the bar ends must be done. Next, the stripper guides are adjusted to touch the roll grooves lightly to prevent any gap between stripper end and the roll. The stripper guide should be tightly fitted into the exit guide box to prevent it getting knocked out during rolling.
- The side guide is also checked for alignment with respect to the pass. The previous pass is checked for the above factors and if in doubt one more sample is passed between the previous stand rolls and the resultant bar is passed through the present stand at which the cobble occurred. If there is nothing further to be done, the next bar is taken for rolling. If the bar passes through smoothly beyond the finishing pass the unit weights of the front and back ends are taken.
- If found OK then the rolling is allowed to continue.

2.4.4 Measures to be adopted for increasing Mill Utilization

- The factors that reduce the mill utilization are:
 - Idle time between two consecutive bars
 - Misrolls
 - Breakdowns
- The measures to be adopted for increasing the Mill utilization are :
 - The idle time between two consecutive bars can occur if there is a late discharge of the next billet from the Reheating Furnace, bow shape of the billets during rolling restricting the entry of the bar into the roll pass and should be prevented. Bow shape occurs if the billet has not been heated uniformly resulting in more elongation on the higher temperature side and less elongation on the opposite side. As a remedial measure the ejector operator should utilize the time between two discharges for getting the next billet in line with the ejector for pushing the billet the instant he gets the signal from the Rolling Mills pulpit operator. The pulpit operator should take into consideration the time lag between the signal and the actual receipt of the billet at the roughing stand and give the signal sufficiently early to ensure the billet reaches the stand the instant the Mill is free to accept the next bar.

- Misrolls should be prevented by proper setting of the Mill & rolling supervision.
- The Mill breakdowns should be minimum which can be achieved by proper operation & adopting Preventive and Predictive maintenance practices of all critical equipments & parts of Rolling Mill as described in the SMP-Base Document.

2.5 STANDARD OPERATING PARAMETERS & INSTRUCTIONS FOR CRITICAL MILL EQUIPMENTS/PARTS

2.5.1 Gear Box & Pinion Stand

- Ensure proper working of the Centralised Lubrication System i.e. the oil pressure at gearbox oil header is 2.5kg/cm^2 and oil temperature $<65^\circ\text{C}$.
- Ensure that oil level in Gearbox/Pinion Stand is such so that $1/4^{\text{th}}$ of the gear dip in oil.
- Check the motor for free rotation by inching the motor and checking the current drawn.
- The diameter of the rolls to be used in the Mill should be such that the inclination of the connecting spindles should not exceed 22.5° , which will happen if the roll diameters are too low or too high.

2.5.2 Spindles

- The spindles are connecting shafts between the pinion housing gears and rolls. At either end of each spindle there is an universal coupling, wobbler coupling, slipper pad coupling or knuckle type coupling. The most common coupling presently in use is the wobbler type coupling, but it is advised that universal type coupling, though costlier, be used for its higher transmission efficiency. The Maintenance on these couplings is also only 30% that of wobbler couplings. Wobbler couplings are efficient if the mating splines are spherical to cater to the height difference between drive end and driven end, but the wobbler couplings in use are straight splines and cannot take the misalignment. Standard Operating & Maintenance Practices dictate that a notch be provided on the spindle to ensure breakage in case of heavy overload of more than 400% caused by collaring which otherwise would have caused roll or gear breakage which are much more expensive to replace and more downtime in the Mill.

- A reinforced spindle guard should be installed to prevent breaking spindles or loose bolts/ covers etc from flying out on breakage.

2.5.3 Mill Housing

- The Mill Housing is normally a Cast Steel structure and the dimensions of mill housing should be such that the required roll barrel length can be accommodated in between the stand posts. The width of the opening of the stand posts should be such that antifriction bearing chocks can be easily fitted.
- After the rolls are assembled in their chocks they are lowered into the stand and will slide down to the bottom-most position resting on the bottom breaker blocks. The top covers of the stand posts carry the screw and nut arrangement for adjusting the roll pass height.
- The side window clamps have jacking screws to move the bearing housings laterally to centre the roll pass with the corresponding pass on the middle roll.

2.5.4 Roller Tables, Tilting, Y-Roller Tables

- There are Roller Tables in front of the Cross Country Mill stands to transport the Billet/Bar to and from the Mill Stands.
- In order for the bar to enter into the pass between the middle and top rolls the bar has to be lifted. The lifting is achieved by (a) Tilting roller table or (b) Y-Tables or (c) Manually if the Bars are not heavy. In Tilting tables the whole roller table is lifted at its front end and the table is hinged/Pivoted at the rear of the table. The lifting is achieved by either a pneumatic cylinder or by a motorized winch moving on a C- structure installed in front of the stand.
- The weight of the table is counterbalanced through mechanical linkages and weights so that the cylinder or winch will have to use only minimal incremental load.
- The lifting operations of the table are controlled by the pulpit operator at the stand.
- The height of lift is adjusted by means of limit /proximity switches and depends on the diameter of the main rolls in the stand.

2.5.5 Wall -Tilters

- This is a series of exit guides that receive the bar from between the top and middle rolls and the bar falls down to the roller table in between the entry guide leading to the next pass on the bottom-middle rolls after turning through 90°. The wall tilter bar supporting surface is not exactly horizontal but slightly skewed to enable the bar to slide downwards by the time the bar is released from the pass. With this arrangement there is no need for two tongsmen on the ingoing side, which would otherwise have been necessary to hold, tilt and guide the bar into the roll pass. Two tongsmen are stationed on the outgoing side standing on the tilting table to guide the bar into the pass between middle and top rolls. Two men are required to take care of the rolling of two bars at a time in the Mill. If Y-tables are used then it is not necessary to have tilting tables as the bar rides up the ramp table into the pass between middle and top rolls.

2.5.6 Front & Back End Cropping Shears

- The approach to the Swivel type Rotary Shears is sensed through photo-electric cells /a proximity switch, which actuates through a timer, the pneumatic movement of the shear, into the rolling line and cuts the front end. The same system is followed for the rear end cutting of the bar.
- While cutting the ends at the crop end Shears, it is very essential to cut as short a length as possible as the one major factor affecting mill yield is the weight of crop ends.
- For this it is necessary to collect the crop ends from the collecting bin and see the extent of split over a Ten crop ends and determining the number of pieces that exceeded 200mm, the number of pieces that exceeded 150mm, the number of pieces that exceeded 100mm etc. If only 10% of the crops exceeded the 200mm limit, and 20% of the crops exceeded 150mm then it must be ensured that the crop ends do not exceed 150mm.

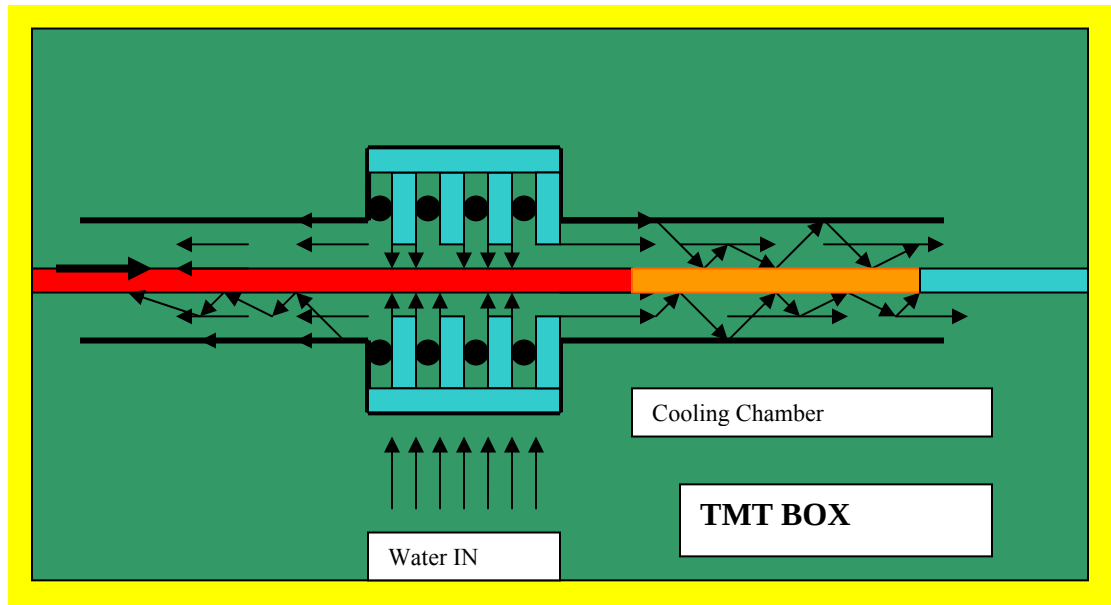
2.5.7 Thermo-mechanical Treatment of Steel (TMT) System

- The TMT System is for martempering the Steel Bar through controlled cooling of the bar through water spraying under pressure as shown at **Figure 2.7.** after applying a mechanical force or draft of 23% or more when the bar is between 850 and 900°C.

- The bar is maintained at a temperature of 850°C at the entry to the finishing mill and given a press of 23% reduction taken into the TMT Cooling water box, cooled to around less than $350 - 500^{\circ}\text{C}$ at 10m length from the TMT Box, at the cooling bed. The strength of TMT bar is codified as Fe 500 (the best and highest quality), Fe 450 etc.
- An on-line infrared pyrometer is provided just before the TMT Box which senses the temperature of the bar at the entry to the cooling box and the temperature controller cum indicator sends the signal to the water flow controller which controls the flow of water in the box appropriately at a pressure of about 5 kg/cm^2 .

Figure 2.7: TMT Box System





2.5.8 Cooling Beds

- Majority of the SRRM units have W-channel cooling bed where the Hot bars are manually cleared after cooling and cut to desired lengths. At a time five skilled persons on an average are needed for clearing the cooled bars on the bed during rolling, two at either end of the cooling bed and one reliever.
- Twin Channel cooling beds have a pneumatically operated deflector plate at the beginning of the cooling bed which directs the bar either into the left hand side channel or into the RH side channel alternately. This is installed for high speed mills. The bars are transported laterally along the cooling bed by serrated walking beams which ensure that the bars remain straight during cooling. The bars are then deposited on to a roller table, which collect ten bars at a time and takes them to a cold shear to cut them to fixed lengths if so required by the customer.

2.5.9 Hydraulic & Pneumatic Systems

- In the Hydraulic systems the important parameters to be observed are :
 - The pressure being developed by the Pumps from the pressure gauge & it should not be less than 80 percent of the rated value.
 - In Pneumatic Systems, the Air compressors are important and the pressure developed by the compressors should be observed from

installed pressure gauge. In addition, the pressure at the Pneumatic System should be as per the rating requirement and normally this is around 4 kg/cm².

2.5.10 Mill Electrics; Power Supply & Distribution System

➤ Mill Electrics

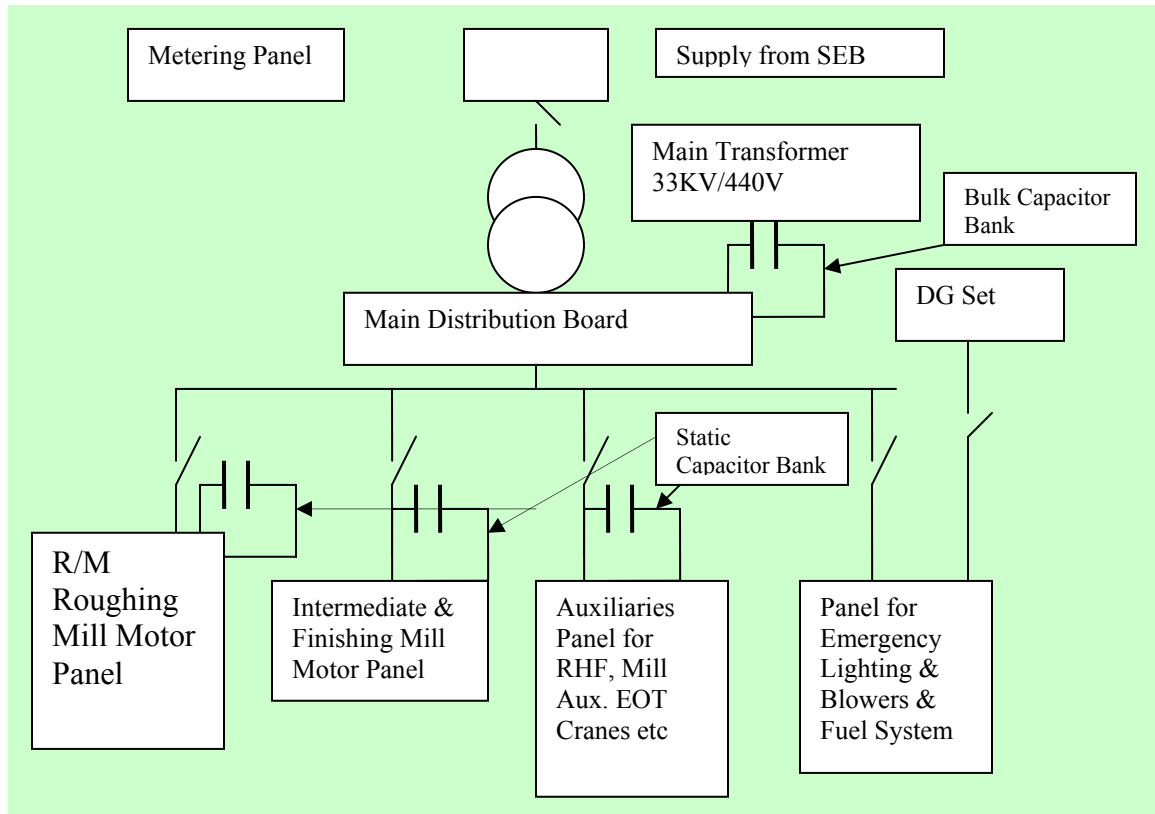
- Includes the following Main Mill Drive Motors for 15 TPH capacity RM:
 - 1350 HP, 760 RPM squirrel cage slip Ring AC Induction Motor to drive Roughing Mill.
 - 850 HP squirrel cage slip Ring AC Induction Motor to drive Intermediate Mill.
 - 450 HP slip Ring AC Induction Motor to drive Finishing Mill.
 - 500 HP capacity (2x 250 HP) DC Motors/VVFD Motors to drive the Continuous Mill Stands.
- It is essential that all the Electric Motors run efficiently. Measurement/monitoring of Motor current, RPM, power consumed through Motor Control Panel/through instruments is essential to ensure lower specific power Consumption in the Rolling Mill.
- The Rolling Mill Motors & Drive system are designed to take 250% instantaneous loading & 150% overloading for 6 seconds.
- Starting the Mill Motor
 - The Mill motors have to overcome a huge inertia at the time it is started. The motor starter used in the Rolling Mills is a liquid starter filled with water and salt solution to reduce the pH value. This acts as a resistance which cuts out as the shunt is raised from the liquid by a hand operated mechanism. As the motor picks up speed the resistance can be cut off faster.
 - Inching operation is carried out by push-button operation from the operator's desk.
- In addition there are Auxiliary Motors i.e. Pinch Roll Motor, Shear Motors, Roller Tables motors, EoT crane Motors etc.

➤ Power Supply & Distribution System

- The 3 Phase power supply at 33/ 11/ 6.6KV is received from the State Electricity Board (SEB) and distributed within the factory by the Units. The incoming power supply is first metered by the SEB before being taken into the system's step-down transformers.
- The Power Lines are isolated through an OCB and then led to a step down Transformer. The Transformer is normally about 1000KVA at 440 V load and all utilities and lighting loads are sourced from this.
- The main mill motors are sourced from another step down transformer of about 5000 KVA and stepped down from 11KV to 440V. The power cables are taken from the transformer into a Main Distribution Board, which in turn has leads to all auxiliary drives and their individual control panels.
- In addition HT Capacitor Banks- 550 KVAR are installed to control PF of main drive Motors of Roughing, Intermediate, Finishing Mills etc. to achieve near Unity Power Factor.
- POWER FACTOR: The capacitors are connected in automatic mode where they switch in and switch out in increments depending on the requirement of the motor to which they are connected in order that the power factor is close to unity at all times. While most of the Units have connected their PF Improvement capacitors in series with the main mill motor drives, when the motor takes overload of about 250 -300% the power factor dips below 0.7, because the capacitors were not designed to improve PF at such high loading. This can be remedied by putting additional capacitors in two increments of $(0.2 \times \text{Motor HP})$ Kvar with separate controls so that the 1st lot of additional capacitors switch in once the loading crosses 150% rated capacity and the 2nd lot of capacitors switch in when the load exceeds 200% rated capacity of motor and vice versa, i.e. the 2nd lot of capacitors switch out when the motor load dips below 199% rated capacity and the 1st lot of capacitors switch out when the motor load goes below 150% rated capacity. In this way the PF can be maintained at near unity under all conditions.

- The Schematic Diagram of a typical Power Supply & Distribution system for SRRM unit is illustrated at **Figure 2.8**.

Figure 2.8: Schematic Diagram of a typical Power Supply & Distribution System for SRRM unit.



- **Cable Routing:** The cables from the sub station to the individual panels at the user points are taken through outdoor trenches for the lengths laid outside and through indoor trenches when laid indoors. If the cable has to cross the shopfloor then it is advisable to take the cables at the roof truss level to avoid damage due to accidental falling of hot steel traveling at high speed. Moreover the presence of cable trench covers on the shop floor are permanent safety hazards both to the cable and to the people working in the area. The sizing of the cables should be adequate to satisfy the capability of carrying the current that would flow through the cables. Proper earthing pits have to be provided as per the Indian Electricity Rules.

2.5.11 Mill Instrumentation & Control System including PC-PLC system

➤ Control Panels

- In the Rolling Mill there are several Control panels or load distribution centres. Each panel is energized through cables drawn from the Main Distribution Board just after the receiving transformer.
- The main Control Panels include:
 - Roughing Mill Motor Panel
 - Intermediate & Finishing Mill Motor Panel
 - Auxiliaries panel for EOT Cranes, other Motors.
- The Instruments of the Control Panel are recalibrated once a year to ensure their proper functioning.
- Each Panel is fitted with Instruments such as Ammeter, Voltmeter, Power Meter, Power Factor Meter, KVAR meter, RPM Meter for VFD Drives wherever applicable.

➤ Process control/ Programmable Logic Control (PC/PLC) for Rolling Mills

1. There are three locations where automation is introduced in the TMT Rolling Mills:
 - **End crop shears:** The sensor is a photoelectric cell placed across the path of the bar to the shear in the intermediate Mill. The cell itself is located behind the sideguard of the trough through which the bar passes. The light from the cell is focused on a receptor on the opposite side of the trough and behind the slot in the sideguard. When the bar passes through, the light ray is cut and the receptor sends a signal to the actuator valve of the pneumatic cylinder which operates and brings the shear in line with the bar and cuts the front 6-8" of the bar and returns to its original position. The same thing happens when the rear end of the bar crosses the photocell the light ray path is now clear and this is sensed by the receptor and the same procedure as above now is applied to the other end crop shear, which cuts 6-8" of the rear end of the same bar. Due to the automation of the Crop shears, the crops are reduced and Mill yield increases by about 0.5%.
 - **TMT Water Box:** An infrared temperature sensor is installed at the entry of the TMT Box, which senses the temperature of the bar at the entry to the TMT Box and operates the water spray valve actuators across the length of the box. The amount of valve opening depends on the temperature of the entering bar and the rate at which the martensite tempering takes place depends on the rate and quantity of water flow.

- **Flying Shear:** The third location for an automatic system is for **actuation of the flying shear**. The flying shear gives the first cut when the bar reaches a preset length which is detected by a photoelectric cell or through a timer which is calibrated to cut at periodic intervals equal lengths of bar as per marketing instructions. The periodicity of cut is determined by the linear speed of the bar and depends on the finished bar sectional dimension.
- 2. **Future Automation:** It is advised to locate a sensor to detect the exit of the bar from the third pass of the Roughing Mill 1st Stand. This sensor will trigger a bell to alert the ejector operator to eject the ingot/ billet from the furnace to the rotating table at exit of the RHF.

2.5.12 Centralised Oil Lubrication & Greasing System

➤ Centralized Oil Lubrication System (COLS)

- Each Stand group like roughing, intermediate (with or without the speed increaser), finishing mill train has its individual Drive Motor, a Reduction g/box, a pinion housing g/box, which has Centralized Oil Lubrication system to take care of proper lubrication of the gears in the g/boxes.
- The pressure of the oil lubrication pumps is set to develop 2.5 kg/cm² at the gearbox oil header. The temperature is not > 65DegC
- The oil is a circulating oil from any Standard Oil Company like Indian Oil, BP, HP etc. The oil should have good viscosity index, good emulsion & detergent properties and should easily separate out the solid particulate matter carried by the oil.
- The system has two sets of filters, a) coarse filter elements and b) fine filter elements to filter the oil before it is pumped to the g/box, which should remain clean.
- The viscosity of the Lubricating oil decreases with temperature i.e. viscosity (mm²/sec) reduces to one third with increase in operating temperature from 400C to 700C, hence the heated oil is cooled to near ambient temperature in shell & tube type Heat Exchanger before recirculation.

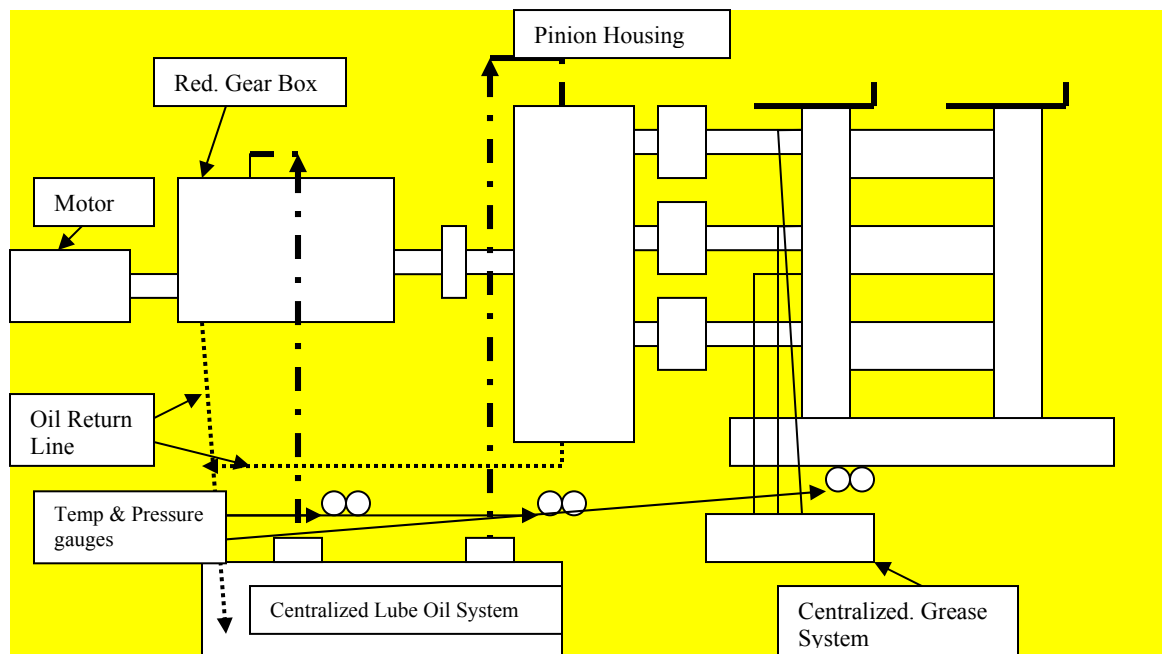
➤ Centralized Grease Lubrication Systems (CGLS)

- For automatic lubrication of the antifriction bearings of the main mill rolls, tilting table linkages, roller shaft bearings and screwdowndown mechanisms etc many Mills have started fitting (CGLS). The system basically consists of a grease reservoir from which two reciprocating pumps pump the grease to the end user points firstly through one pipe. From this pipe the grease enters a manifold with a plunger which is pushed upward by the grease pressure. The grease from the top of the

plunger enters the bearing housing greasing opening. Once all the plungers have been pushed upwards, a pressure switch switches off the pump. Through a timer of 20 minute intervals the other pump starts and pumps the grease through the second pipe which enters the manifold and pushes the plunger downwards. A metered quantity of grease enters the bearing housing. As in the previous case, when all the plungers have been pushed downwards, the pressure switch operates to switch off the pump. When the pump has operated for 20 minutes and the pressure switch not operate for want of a signal, a siren is blown to alert the mechanical staff to attend to the CGLS.

- Schematic Diagram of a Typical Centralised Oil & Grease lubrication System
- The Schematic Diagram of a Typical Centralised Oil & Grease lubrication System at stand Reduction G/box, Pinion Housing, Stand Roll bearings is shown at **Figure 2.9**.

Figure 2.9: Centralised Oil & Grease Lubrication System at Stand Reduction G/Box, Pinion Housing, Stand Roll Bearings

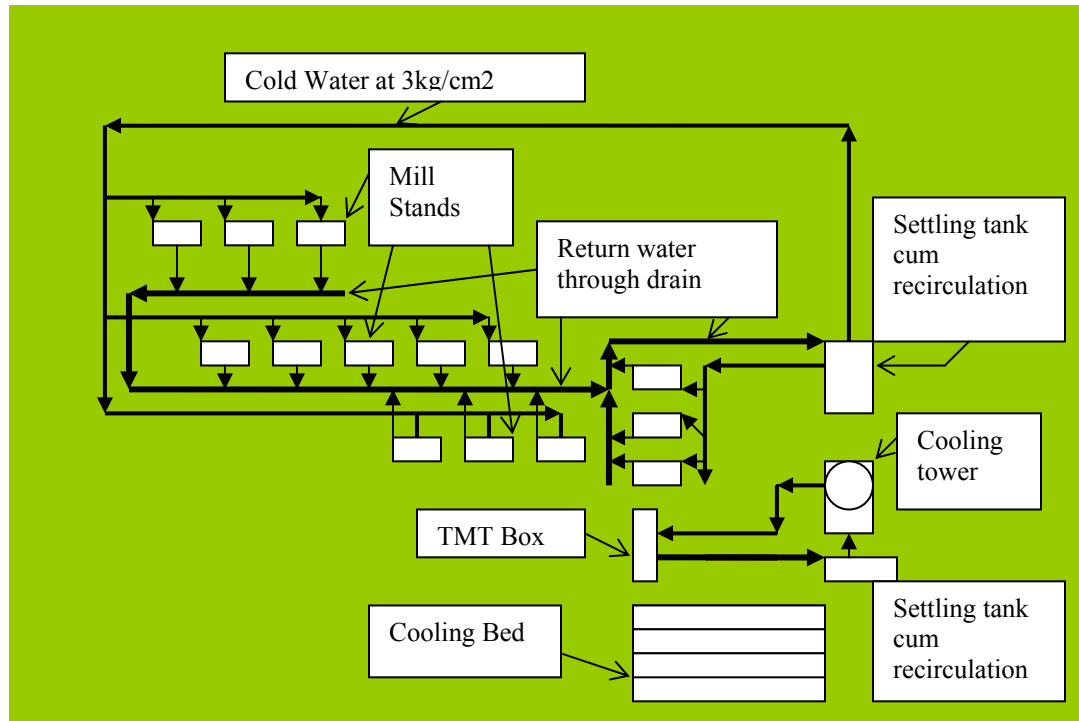


2.5.13 Mill Cooling Water System

- Cooling Water is needed primarily to Cool the Rolls, cool the oil in the centralized lubrication system, blade cooling in crop shears and in TMT Box.

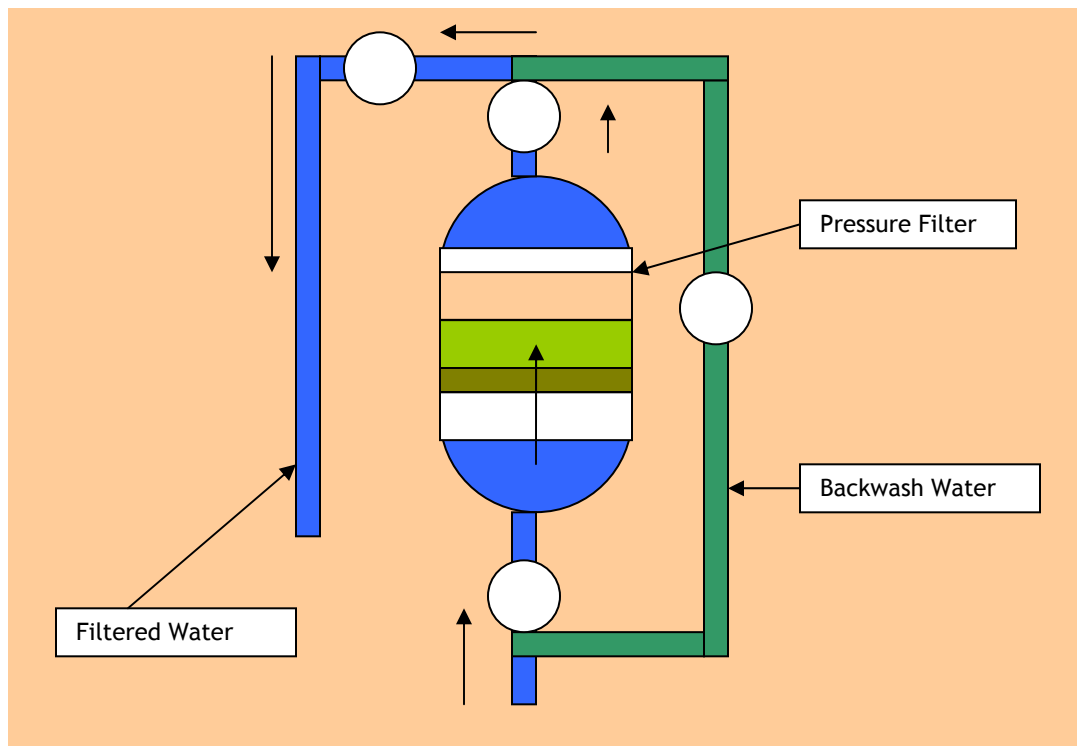
- The Schematic Diagram of the closed circuit Mill Cooling water system is shown at **Figure 2.10**.

Figure 2.10: Schematic Diagram of Rolling Mill Cooling Water System & Piping



- There are Separate Settling Tanks & Recirculation Systems for Mill stands and TMT Box. The water returns back to these tanks through return water drains.
- The Cooling Water pressure in Mill Stands is around 3kg/cm² and in TMT around 5 kg/cm², hence the Mill Cooling water system discharge pressure should be around 7.5-10kg/cm².
- The Hot Water is re-circulated back to the Cooling tower that cools it to around 30⁰C.
- It is advisable to install pressure filters at the recirculation tanks to filter out all un-dissolved solids so that the nozzle sprays for roll cooling do not get choked (Shown at **Figure 2.11**). The pressure filters need to be sized to handle only 10% of the total recirculation water. This is known as side-stream filtration system.

Figure 2.11: Schematic Diagram -Cooling Water Filtration System



- The flow rate of cooling water is also very important and needs to be monitored from the flow meters installed on the Cooling Water delivery side. For a 15TPH capacity Cross Country RM the total water requirement for cooling of Rolls & TMT Box is around 250 M³/hr in a closed loop system with cooling tower.

2.6 ROLLS & ROLL MANAGEMENT

2.6.1 Different Grades of Rolls & Typical Sizes used

- The Rolls being used by the RM are available in three materials:
 - Alloy Steel : EN8, EN9, EN16, EN19, EN24, EN31B etc
 - SG Alloy Cast Iron: Bainitic SG Iron, SG- Pearlitic
 - Cast Iron: Indefinite chill CI, Double Pour Indefinite chill CI
- The diameter of the Rolls and also their speed (RPM) vary from Mill to Mill and is based on the type & size of section being rolled, stand where rolls are mounted etc.
- The stand wise Roll diameter, speed & roll grade for a typical 12 stand RM in SRRM sector are provided at **Table 2.3**.

Table 2.3
Stand wise Roll diameter, Speed & Grade for a typical 12-Stand Rolling Mill in the
SRRM Sector

MILL STAND	MAX. ROLL DIA (mm)	RPM	ROLL QUALITY/GRADE	
			TYPE	HARDNESS(⁰ SHORE)
R/M-1	420	120	Alloy Steel.	40-50
R/M-2	430	120	Indefinite Chill CI	45-50
R/M-3	435	120	Indefinite Chill CI	45-50 □
R/M-4	435	120	Indefinite Chill CI	45-50
I/M-1	270	330	Indefinite Chill CI	55-60
I/M-2	280	330	Indefinite Chill CI	55-60
I/M-3	290	330	Indefinite Chill CI	55-60
I/M-4	310	400	Double Pour Indefinite Chill CI	60-65
I/M-5	330	400	Double Pour Indefinite Chill CI	60-65
F/M-1	280	470	Double Pour Indefinite Chill CI	70-75 □
F/M-2	290	470	Double Pour Indefinite Chill CI	70-75
F/M-3	315	470	Double Pour Indefinite Chill CI	70-75 □

- **SG Alloy Cast Iron:** This is a combination of high strength SG (Spheroidal Graphite) Iron material with inherent properties of Alloy CI and is used in most Rolling Mills rolls today. One such is the Bainitic SG Iron, which has found universal appeal in Roll Material.
- The properties of Bainitic CI Roll Material are provided at **Table 2.4.**

Table 2.4
Hardness, Chemical properties & UTS of Bainitic CI Roll Material

	Chemical composition (%)								
Hardness (⁰ Shore)	C	Mn	Si	P	S	Ni	Cr	Mo	UTS (kgmf/mm2)
40 – 50	3.0-3.3	0.3-0.5	1.8-2.3	0.08 Max	0.015 Max	1.5-3.0	0.2 Max	0.8-1.1	60 -80
50 – 60	3.0-3.3	0.3-0.5	1.8-2.3	0.08 Max	0.015 Max	1.5-3.0	0.5 Max	0.8-1.1	55 -75
60 – 70				0.08 Max	0.015 Max	2.0-3.5	0.3-0.8	0.8-1.1	50 -70
70 – 80				0.08 Max	0.015 Max	2.5-4.5	0.3-0.8	0.8-1.1	50 -70

2.6.2 Method of Selection of Rolls

- Rolls are the main working tools of the Rolling Mill and right selection of Roll has a tremendous impact not only on the availability & productivity of the RM but also on the Product quality & Mill economics as Rolls constitute about 40 percent Cost of Consumables.
- The Material of Roll, its Grade (Hardness) & Size should carefully be selected based on factors such as:
 - Product
 - Product/Section to be rolled i.e. Heavy Section; Medium Section; Light Section; Rod; Grade of Material being rolled i.e. MS, Alloy steel etc & its Hardness.
 - Location of Roll
 - The Mill Stand where the Roll will be used/mounted in the RM i.e. Roughing, Intermediate, Finishing etc as different Roll Surface Hardness is required at different Mill Stands based on Drafting, Rolling Speeds etc. The Hardness of Rolls required in various Mill stand is:

- Roughing Mill	: 45-55 ⁰ Shore
- Intermediate Mill	: 55-60 ⁰ Shore
- Finishing Mill	: 65-75 ⁰ Shore
 - Roll Diameter Selection
 - The maximum and minimum diameters of rolls that can be used is dependent on the centre to centre distance of the pinion housing and the diameter of the spindle coupling covers, so that the outward angle of the spindles is not greater than recommended by the equipment supplier, nor should the spindle couplings touch each other on the roll side due to low diameter of the rolls. Another important factor of roll life is that the depth of hardness is limited upto a certain depth depending on the chill given to the casting. The core would be soft and can not be used for rolling.
 - Roll Pass Design/Schedule

The percentage reduction in each pass & stand has a tremendous effect on Roll behaviour & roll material/Grade selection as high reduction in stand requires High strength rolls sacrificing wear property & low reduction in stand allows Low Strength High wear resistance rolls.
 - Roll Cooling System

The roll grade selection depends on the type of water cooling system available in the RM.

- Gap Time Between Bars

The gap between successive rolling stocks influences the roll temperature, which in turn affects the roll behaviour due to thermal stress. So suitable grade of roll, which has high thermal fatigue resistance has to be selected.

- Descaling Practices

The oxide or rolled in scale on the bar will have a great effect on roll life, high wear and pits denoting rolled in scale. So roll quality has to be selected depending on the configuration and Systems available in the Mills. High Strength and High wear resistance combination for Roll will reduce the above effect.

- Past Problems & their Analysis

Selection of correct roll grades depends to a great extent on the data/analysis of problems faced in the mills (firecrack, slippage, pitting, spalling, breakage, rolled in scale). The past experience should be clearly addressed while procuring the new rolls.

- Roll Changing Schedule

Roll changing schedule/campaign life has a bearing on roll life. Correct roll grade selection has to be done depending on campaign life, the available barrel lengths. Depending upon the campaigns the rolls can be dressed accordingly (single or composite rolls).

- Roll Dressing

Under dressing & overdressing of roll surface between campaigns influence the roll life. Proper selection of rolls material (chill hardness etc.) can enhance the roll life.

➤ Latest Roll Grades, their Benefits & Usage

Some of the new grades of the rolls are listed below which could beneficially be used by SRRM in place of Conventional Rolls:

- Dispersed Carbide Indefinite Chilled Roll

- Application: Finishing Mill
- Benefits:
 - a) Increased life of about 30% to 40% over normal indefinite chilled roll
 - b) Higher campaign life

- Dispersed Carbide Nodular Iron Roll

- Application: Bar & Rod Mill
- Benefits:
 - a) Increased life of about 30 to 40% over normal Nodular Iron
 - b) Resistant to fire crack

- c) Higher campaign life
- High Speed Steel Roll
This is the latest developed roll material. Rolling mill all over the world have started using this grade getting wide benefits.
 - Application:
 - a) Bar & Rod mill (both intermediate & finishing stand).
 - b) Small section like channel & Angle mill.
 - Benefits
 - a) More than three times higher roll life over conventional rolls (cost is only two & half times-over conventional grades)
 - b) More than two times campaign life over conventional grades causing higher productivity & reduced down time.
 - c) Less dressing which reduces the load on turning/grinding machine
 - d) Excellent surface finish & high dimensional accuracy of the finished product.
- Soft Annealed Nodular Iron
 - Application: Roughing stand for bar & Rod mill, Structural mill
 - Benefits:
 - a) High fire crack resistance than normal nodular iron.
 - b) High wear resistance by about 20-30% over nodular iron.
 - c) Lower amount of dressing between campaign (about half than that of normal nodular iron).

2.6.3 Rolls Inventory

- Rolls Inventory is very important as Rolls could be damaged/wear out during rolling and need immediate replacement without stopping the rolling mill for long.
- The recommended roll inventory schedule is given at **Table 2.5** for a 9- Stand cross country mill rolling rounds from 57 mm dia upto 8mm dia, squares from 10 sq inch to 30 sq inch and flats from 25x6mm to 80x10mm (i.e. 150 mm² to 800 mm²) which shows that for efficient RM working for a usage of 46 Rolls a Corresponding 33 Spare Rolls need to be maintained.

Table 2.5
Recommended Roll Inventory for a 9-Stand Rolling Mill

Mill Stand #	For Profile/ Section	Number of drawings	Drawing No (Code)	No. of roll sets	No. of rolls per set	Total No. of grooved rolls to be maintained	Spare rolls to be maintained
1	Φ < 25 Set 1	1	1.100	1	3	3	3
	Φ > 25 Set 2	1		1	3	3	
2	Φ	1	2.100	1	4	4	4
	Φ	1	2.101	1	4	4	
3	Φ	1	3.100	1	3	3	8
4	Φ	1	4.100	1	3	3	
5	Φ	1	5.100	1	2	2	
	□	1	5.101	1	4	4	
6	Φ	1	6.100	1	2	2	2
		1	6.101	1	2	2	
7	Φ	1	7.100	1	2	2	2
	□	1	7.101	1	2	2	
8	Φ 282 Set 1	1	8.100	1	2	2	4 Φ 282, 2 Φ 302
	Φ 302 Set 2	1	8.101	1	2	2	
	□	1	8.102	1	2	2	
9	Φ	1	9.100	1	2	2	8
	□	1	9.101	1	2	2	
	□□	1	9.102	1	2	2	
Total				18	46	46	33

Φ – Rounds, ■ squares, — Flats

- Similar Roll inventory models need to be prepared for individual Rolling Mills, depending upon the number of Stands, Rolling Schedule, tonnage rolled per section etc.
- The Rolls should be stacked on Roll Stands Mill Stand wise and be numbered for easy identification.

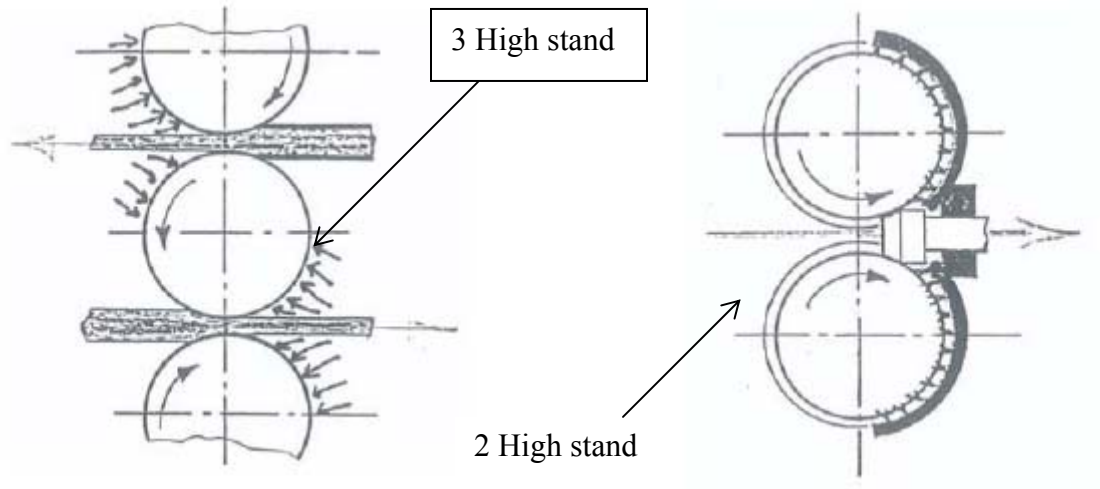
2.6.4 Roll Cooling

- Roll cooling is one of the major parameter for optimizing the roll life in the running mill.

- Major reason for fire crack generation on a roll surface is inadequate and improper roll cooling, which results in deeper cuts during roll turning to remove the firecracks.
- A good roll cooling means roll should be brought to ambient temperature with uniform distribution of temperature throughout the barrel length.
- It is essential that 1/3rd of roll surface at the roll pass should be water cooled.
- The cooling water pipe should be located on the delivery side of the pass and not on the top of the roll barrel. The pressure of the cooling water should be $>2\text{kgs/cm}^2$ and flow should be copious. The water should be less than 50degC and free of suspended solids. A second pressure gauge should be installed at the mill roll housings at each individual stand to monitor that the correct pressure of water is maintained.
- The Cooling Water pipe line should have pressure & temperature gauges to ensure correct values.
- The return water from the scale flumes are generally filled with scale and oil/grease from the roll bearing effluents. The scale is lessened by settling in the settling tank in three stages and the oil /grease is skimmed off by the cross beam located in the tank. A side stream pumping and filtration system should be installed to ensure clean filtered water to the roll cooling water system for longer life of rolls.
- Since the heat is input into the roll only at the roll pass it is inadvisable to bathe the rolls along their entire barrel lengths, but the cooling water should be sprayed through nozzles only on the Roll Pass in use.
- The configuration shown in **Figure 2.12** for cooling the top and bottom rolls of 3Hi-Stand and 2Hi-Stand should be adopted for efficient Roll cooling.
- A straight square cooling water pipe clamping rest bar of 60 mm side is fixed across the roll barrel on the underside of the guide rest bar and clamped to it. The roll cooling pipes as shown in the figure are clamped at the exact centre of the roll pass in use.

- There are two vertical water headers on either end of the roll barrel which feed these roll cooling spray headers through rubber hoses.

Figure 2.12: Recommended Roll Cooling Water Layout



2.6.5 Importance of Roll Speeds in Repeater Rolling

- The linear speed at the centre of any roll pass should be 2% higher than in the previous pass.
- The linear speed of a roll is calculated by πDN where D is Average Roll Diameter in mm and N is roll RPM.
- The linear speed of a Roll of diameter 400 mm rotating at 120 rpm is 151 m/min.
- The succeeding roll pass linear speed should be 148 metres/min or 2% lower.
- If the drive is common, i.e. the RPM is still 120 then the diameter at the next roll pass should be $148 / (120 \times \pi) = 392\text{mm}$.
- If the diameter of the succeeding roll pass is greater than 400mm then there would be tension in the bar and either the bar will snap or the motor will trip due to overload.
- If the diameter of the succeeding roll pass is lesser than 392mm then there would be a long loop, which will be exposed to the atmosphere

until it exits the roll pass. This results in two damaging effects, 1) Excessive scale formation and 2) reduction in bar temperature.

2.6.6 Roll History Card & Roll life

- The Roll History Card for each Roll of the Rolling Mill need to be maintained by the RM Foreman with recorded details as per the Format provided at Table 2.6.
- Based on the Data recorded in the Roll History Card for each Roll, the average life of Roll in terms of Tonnes of Material rolled per mm radius of Roll is calculated and compared with the Roll life specified by the Roll supplier.

Table 2.6
Recommended Format for Roll History Card

1. Roll No.....
2. Chemical Composition (Supplier's T.C)

%C	%Mn	%Si	%CNi	%Cr	%Mo	%S	%P

(Unit's Chemical Analysis)

%C	%Mn	%Si	%CNi	%Cr	%Mo	%S	%P

3. Shore Hardness
4. Roll Date Received.....Supplier.....
5. Roll Date Machined.....MRN No. & Date.....
6. Roll Date Commissioned
7. Physical Condition.....
8. Roll Scheme.....
9. Roll Size..... Type.....
10. Position of Roll.....
11. Stand No.....
12. Set No.....
13. Roll Weight (Recd).....Roll Weight (After Machining).....
14. Roll Machining History

Sr.	Roll Machining Date	Roll Diameter		Production During One Machining	Total Production	Remarks
		Old	New			
1.						
2.						
3.						
4.						

2.6.7 Roll wear & Reclamation

- Roll Wear
 - Higher drafting in Mill Stands results in faster wear out of Roll pass schedule design is necessary.

- Inadequate Roll water cooling leads to fire cracks on the surface, reduced roll life.
- Improper selection of Roll Material/Grade (its Hardness) leads to quick wear out of Rolls, pitting on roll surface.
- Overdressing of rolls, shortens Roll life.
- Higher Roll speed also leads to faster wear of Rolls.
- High & Non-uniform wear of roll is due to defective Roll Material and wrong rolling practices.
- Chipping/Spalling of roll surface due to rolled in Scale particles make the roll weak & ultimately the rolls wear out.
- Many Units rag or bead-weld on the roll passes especially at Roughing Mill Stand#1 to enable higher size of input to be rolled by improved bite angle. The bite angle should ideally not exceed 22° . The welding beads increase the friction between the input bar and the roll pass and drag the bar into the pass. Each bead has an increased bite angle which enables the bar to be dragged in. This is not a healthy practice as it decreases the roll life, induces a lot of stresses in the roll.

➤ **Roll Reclamation**

- Rolls made of alloy steel (Grade En 8, En 9, En 16, En 19, En 24, En 31B) used mainly in the roughing stands, can be reclaimed by welding hard surfacing layers and subsequently annealing the same, either by flame annealing at 6500C or shot peening to relieve the built up stresses. If the welding is extensive the rolls are ground finished on the Lathe by attaching a roll grinder to the tool-post.

2.7 MANUFACTURING BEST PRACTICES

➤ **Customer Sensitive Organization**

The organization must be extremely sensitive to the present and future requirements of the customers. They must be able to anticipate the perceived and unperceived requirements of the customers.

Many a times the requirements of the customers are unarticulated - The success of any organization depends upon their ability to articulate those needs and manufacture them in order to satisfy the customers.

➤ **Philosophy of three joys**

Every one believes in the philosophy of '**Three Joys**'. The product you manufacture must be:-

- Joy for people who produce them
- Joy for people who sell them and finally
- Joy for people who buy them

➤ **Plan for quality**

Product quality must be planned and there should be no short cuts to quality. All problems must be taken as sources of improvement.

➤ **Reliability**

Product reliability must be the topmost priority. Study all factors that can hamper uniformity between products and their '**long term stable performance**'. All safeguards must be taken at the product design stage itself rather than post design corrections - in other words robust design.

➤ **Preventive approach**

Quality is achieved through prevention and not appraisal.

➤ **Zero defects**

Quality standard must be zero defects and not acceptable levels. The products with defects or even hint of defects should never reach the customer.

➤ **All pervasive quality**

Quality must be made a company wide issue and must be an all pervasive influence on the way every aspect of business is conducted.

➤ **Just in Time**

Goods and components must be moved to the **correct and useful place** only at a time when the movement needs to take place

➤ **Mistake proofing**

All sources of error must be eliminated in such a way so that it is impossible to perform the task in the wrong way.

➤ **Competitive Quality**

What ultimately matters is not just quality but how good you are compared to the others in terms of Cost, Quality and Delivery. This requires organizations to constantly innovate and improve. The companies must regularly practice PDCA (Plan, Do, Check, Act) and QFD (Quality Function Deployment) to improve its signal to noise ratio and competitive quality.

➤ **Constantly Monitor Quality Costs**



All efforts must be made to reduce waste and non-value adding activities. Right from the beginning the organizations must invest in preventive costs rather than spending money in appraisal, internal failure and external failure costs.

➤ **New Technology**

The organizations must use the latest technology before the old ones become obsolete.

➤ **Cutting Buffers**

Organizations must reduce inventory, raw material, work in progress. Finished goods, set-up times, time to market and knowledge buffers. Organizations must also cut the authority buffers and empower the work force to develop the effective solutions.

➤ **Process Automation**

The organizations must automate the process engineering process across the organizations. All the routine processes must also be automated

➤ **Supplier - Best practices**

- a. Products are 100 percent correct & reliable.
- b. Deliveries are always on time

- c. Quantities delivered are always correct
- d. Deliveries occur frequently to minimize stock carried by user.
- e. The supplier provides appropriate response to urgent requirements
- f. If something goes wrong, there is total commitment to rectifying it as soon as possible.
- g. Product pricing is competitive.
- h. Invoices and documentation are free from errors.
- i. The supplier is totally open and honest about processes, costs and pricing methods.
- j. The supplier works with the company to continuously improve performance.

CHAPTER 3

MONITORING; MEASUREMENTS & QUALITY CONTROL PROCEDURES

3.1 TEMPERATURE MONITORING & CONTROL

- The ingots/ billets coming out from the RHF are checked for proper soaking and scale covering.
- The soaking of the ingots/billets depends on the duration the ingot/billet is kept at the same temperature for 4 minutes per inch height of ingot/ billet. If the ingot is 4.5" (110mm) cross section, the ingot should be kept at the required rolling temperature for $(4 \times 4.5) = 18$ minutes. Hence the final temperature of 1100°C should be reached when the ingot is at $2.2\text{m} + 1\text{m} = 3.2\text{m}$ from the discharge door centre-line. The end of the heating zone is approx 5m from centre-line of discharge door. It is recommended that the final temperature of 1100°C should be reached soon after the ingot leaves the heating zone or at least 3.2m from discharge door centre-line.
- The temperature of the Ingot/Billet exiting the furnace should be monitored on random basis using Portable Infrared Pyrometer (**shown at Figure 3.1**) to ensure trouble free rolling in the mill. If the temperature of the bars exiting the RHF is below or above than what is needed, the rolling mill operator immediately sounds the RHF operator who takes the corrective measures to regulate the RHF combustion system accordingly. Both the temperature controller cum indicators are adjusted to regulate the fuel supply as the temperature at the end of heating zone/ soaking zone approaches 1100°C , cutting off the supply completely when the temperature crosses 1200°C . The fuel supply is resumed when the temperature reaches 1080°C .

Figure 3.1: Portable High Temperature Infrared Pyrometer



- In PLC Controlled Reheating Furnaces there is an infrared billet counter which can be programmed to read the temperature of the exiting bar also in addition to counting of the bars and the period **elapsed** between ingots/ billets. This counter is fitted at the corner of the furnace end wall and discharged side of the furnace at roof height and pointing towards the discharged ingots/ billets. As soon as the infrared light pointer touches the side of the ingot/ billet the temperature gets automatically recorded and counted, both of which readings are displayed in the digital readout.
- The temperature of the Rolling stock in the Rolling Mill should be monitored on a random basis with the hand held infrared pyrometer described above, at the following three critical points and controlled either at the RHF level or at the RM level by reducing rolling time:
 - As the bar leaves the final pass of Roughing Mill
 - As the bar leaves the final pass of Intermediate Mill
 - As the bar leaves the final pass of Finishing Mill
- There is an infrared pyrometer installed at the entry to the TMT water quenching box which notes the temperature of the bar at this location and the reading is indicated on the mimic panel of the operator. Another infrared pyrometer is installed at 10m distance from the end of the TMT box, to ensure that the temperature is

around 350⁰C. Some TMT boxes automatically take the signal of this temperature from the temperature indicator cum controller and vary the flow and pressure of the water to quench the bar in the martempering process. The water valves are motorized and can be infinitely controlled from zero to maximum by the impulse signal received from the temperature controller cum indicator.

- It must be noted that the temperature of the bar increases during the rolling process due to the work being done on it by the rolling forces. The temperature drops when the bar has to travel long distances between stands or is made to wait at the stands and between the stands, i.e. there is a lot of idle time and is exposed to open air. It is at this time that the scaling takes place rapidly on the bar and this should be prevented. The drop in temperature of cross country manually operated Mills is higher than that obtaining in repeater operated mills. Even in repeater installed mills the temperature drop depends on the length of loops between passes. The temperature drop is minimal in continuous Mills, in fact the temperature increases as it passes through succeeding mill stands.
- The size of the bars, both for dimension as well as for weight per metre, should be checked at least once every hour at the front end and back end of the finished bar by cutting a one foot sample. The hot dimension is measured using 'go-no go' gauge templates. The dimensions like diameter, thickness are checked using vernier calipers. Rolled products from the section mills like I-Beams, C-Channels etc are measured using templates prepared to accurate dimensions. If there is a large variation between the weight per metre at front and back ends, then it means that there has been a sizeable drop in temperature of the bar due to the above mentioned reasons, which should be immediately corrected.
- The Surface temperature of the Rolls **at the roll pass** also needs to be monitored using portable pyrometers to ensure proper Roll Cooling.
- The temperature of other systems like Lubricating oil, Cooling water being recirculated in closed loop systems also need to be monitored through installed Temperature gauges & controlled by adjusting the

pressure and flow of water to be as close as possible to near Ambient Temperature ($<40^{\circ}\text{C}$). In case of excess water temperature an aeration system can be introduced by blowing air through nozzles on a long pipe under the water. The best practice would be to install a cooling tower and a pressure filter in the recirculation line to keep the water cool & free of suspended particles

3.2 SECTION MONITORING & CONTROL

- The Rolling Mills Shift-in-Charge carries with him a Roll Pass Schedule for the Section being rolled. At the location where stage inspection is carried out appropriate templates, vernier calipers, micrometers, and go/ no-go gauges are kept. One piece of 1st long is cut from the finished bar, once every hour and the dimensions achieved are minutely checked using vernier calipers and/or templates and gauges.
 - In case the section is displaced between top and bottom halves the roll squaring is carried out and the window clamp bolts fully tightened.
 - In section rolling like when rolling angles it is found that one leg of the angle is heavier or larger than the other leg, then the entry guide at the pre-forming pass at Stand#1 is re-aligned and the fixing bolts fully tightened so that the bar enters the pass centrally.
 - In case there is fin-formation along the parting layer between top and bottom rolls, this means that the material fed is in excess of what the pass can take and the preoval pass is made smaller or this pass is opened a little.

3.3 MONITORING & CALCULATION OF MILL UTILIZATION

- Mill utilization is the ratio of the total time the mill was utilized for actual rolling to the total time the mill was available for rolling.
- The mill utilization is adversely affected by various parameters like misrolls, cobbles, breakdown of RHF due to which the RM has to be stopped, electrical power outages, breakdown itself of the RM and all these factors occurring during the shift are recorded by the RM operator in the Log Book

- For example for a Mill rolling capacity of 15 Tph and a Reheating Furnace capacity of 16 Tph, 18 hrs rolling per day and 300 days per year the Mill utilization can be calculated as follows and this could be adopted by the individuals SRRM units for calculating their mill utilization at the end of each day:
 - Daily Mill rolling capacity: 270 t
 - Available hours for rolling: 18 hrs
 - Idle time between passes: 3 hrs
 - Mill setting time: 1hr (planned)
 - Actual available hours for rolling: $18 - 1 = 17$ hrs
 - Mill utilization time: 14 hrs
 - Mill utilization %: $14 \times 100 / 17 = 82.35\%$
 - Actual production tonnage: $270 \times 82.35 = 222$ tpd or 12.4 tph as against 15 tph rated capacity.
- The suggested Format for recording Mill Utilization parameters is provided on a Daily Basis is provided at **Table 3.1**.

Table 3.1
Format for Recording Mill Utilization Parameters

Available hours for rolling (hrs)	Idle time between passes (hrs)	Mill Setting Time (hr)	Mill utilization (%)

3.4 SCALE LOSS DETERMINATION (RHF, RM, TOTAL)

- Scale Loss is a serious issue for all Rolling Mills in the SRRM Sector. While the larger enterprises have restricted their total mill scale loss to less than 1% the average scale loss in the SME- SRRM Sector is greater than 3% and this reduced the Mill yield considerably.

➤ The overall scale loss in the SRRM is determined on a Daily Basis as follows:

- Two ingots/ billets are weighed in a weighing scale with ± 0.25 kg accuracy and charged into the Reheating Furnace, duly marking the same with bricks.
- After two hours two more ingots/ billets are similarly weighed and charged.
- When the above ingots/ billets reach the discharge end, one of the two ingots/ billets is kept outside near the furnace and cold water is poured over it to cool it sufficiently (below 700°C) to stop further scale formation, or if the ingots are alloy steel then the ingots are buried under dry sand heap.
- The second ingot/ billet is sent through the rolling process to the finishing end. On the way the end cuts from this ingot/ billet are collected, together with any misrolls if any and weighed.
- The finished bar is weighed at the same weighing scale as before. The difference between the initial weight of the ingot/ billet, the finished weight and end cuts plus Misrolls weight gives the total scaling weight.
- The other ingot/ billet which was segregated after the furnace is now cleaned of all the scale sticking to its surface by means of chisel and / or light grinding, and the ingot/ billet is now weighed as before. The difference in the weight before charging and after discharge and cleaning gives the weight of scale formed in the furnace.
- The difference between the total scale formed and the scale formed in RHF gives the weight of scale formation at the Mill.
- The whole process is repeated for the next two weighed ingots/ billets which had been charged into the furnace.
- The Format for recording parameters related to Scale Loss determination is provided at **Table 3.2.**

Table 3.2
Format for Recording Scale Loss determination parameters

[illegible]

3.5 MILL YIELD

- The yield of the rolled bar from Ingot/Billet to finished product is the most important factor that determines the profitability of running the rolling operations.
- The various factors influencing the Mill yield are:
 - a. Scale loss in the Furnace
 - b. Scale loss in the Rolling Mill
 - c. Misrolls
 - d. Short lengths
 - e. End cuts
- For every batch of Rolling, the percentage of total weight of finished products to the total weight of Ingots/Billets charged in the RHF gives the overall Mill Yield. This is normally determined on a daily basis.

3.6 SPECIFIC POWER CONSUMPTION & POWER FACTOR

- Rolling Mill in the SRRM is the major consumer of electricity with an approximate connected load of 5000 HP for 15 TPH capacity.
- Every Motor Control Centre (MCC) Panel for the main drives of Roughing, Intermediate & Finishing Mills must have the following instruments for monitoring purposes:
 - Power Factor Meter
 - Ammeter
 - Voltmeter
 - kWh meter
- The method for calculating the total Power consumed by the Rolling Mill in a Shift is by noting the start & end reading of the RM feeder main Power Meter (Kwh). This divided by the Tonnage rolled in that shift provides the specific Power consumption for the Mill in Kwh/Tonne.
- The power factor meter readings are also recorded on an hourly basis.
- The suggested Format for recording various parameters related to SPC in Mill is provided at **Table 3.3.**

Table 3.3
Format for recording various parameters related to SPC in Mill

Date: _____ Roughing / Intermediate/ Finishing
Mill _____

Hour 1	Pf 2	Amps 3	Volts 4	kWh 5	Production (Tonne) 6	SPC (Kwh/T) 5/6
1 st						
2 nd						
3 rd						

- Mill Motors are designed to withstand overloads upto 3 times its rated capacity for 3 seconds
- The power factor correction capacitors are normally designed and installed based on the full load rating of the Motor. When overloaded, the pf for the motor drops. This is a loss of energy and should be avoided by connecting additional capacitors with automatic cut-off provision that ensures that near unity power factor is achieved all the time at not only the receiving station but also at the individual motors.
- Even though with the installation of capacitors on the HT side will ensure near unity pf it only ensures that the energy taken by the Unit is efficiently utilized and this ensures that there is no penalty imposed, rather a bonus is earned from the grid supplier. But it does not ensure that the power is efficiently utilized downstream at the LT side, which can be ensured only if capacitors are installed on all inductive loads, so that near unity pf is achieved at each major drive.

3.7 QUALITY INSPECTION OF FINISHED PRODUCTS

3.7.1 Visual Inspection

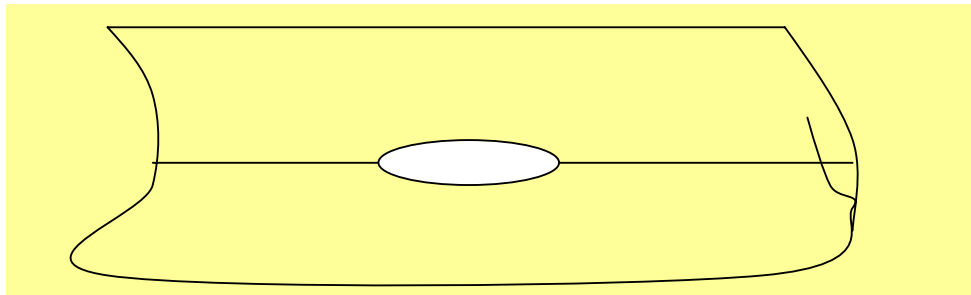
- The visual inspection is done to ensure that the Bar is free of surface defects like Longitudinal cracks openings (shown at **Figure 3.2**) which occur due to tight roller guide & non rotation of rollers of roller guides; from foreign particles embedded in steel or from a seam caused by weld beads. In addition any Cross Cracks (shown at **Figure 3.2**) which are found on edges of

Square Bars or on round bars due to low temperature rolling are also monitored.

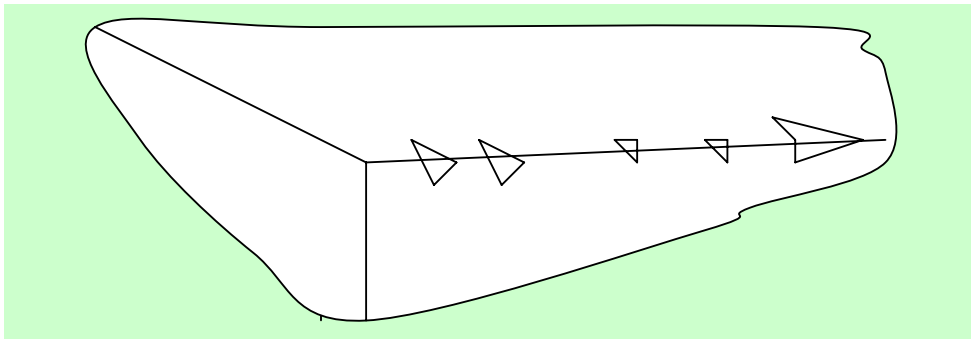
- Visual inspection also shows irregularities in the rolled product, like finning (or whiskers), side displacement caused by shift of the top and middle rolls with respect to each other.
- A slight darkening of the inside of the bar indicates presence of piping which is highlighted by excessive spread at regular intervals, which could cause misrolls and /or splits in the bar.
- The operator should not wait for the prefinishing pass to correct section discrepancies, but bring the correct section from each roll pass to the next pass so that smooth defect free rolling could be achieved and equal loading of the motor is ensured across the different roll passes. For this the temperature should be within the range specified for each section.
- If difficulty is encountered in entry of the bar into the roller guide box it should be immediately corrected by adjusting the roller guide box and/or the previous leading pass to deliver the correct size to the next pass.

Figure 3.2
Surface on Hot Rolled Bars

A) Longitudinal Cracks Opening



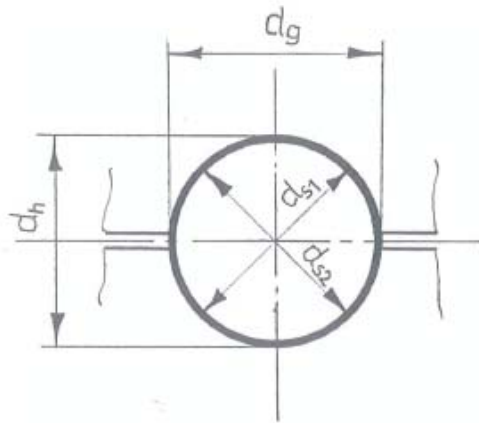
B) Cross Cracks



3.7.2 Profile & Dimensions checking

- The Dimensions Standards for various Hot rolled Alloy steel products are:
 - DIN 1013 for Round Bars
 - a) The Standard covers Permissible Variation in Hot Rolled Round Bar Diameter, Ovality Length, Weight & Straightness
 - b) Measurements to be carried out
 - i) The diameter would be measured at the beginning, middle & Bar end.
 - ii) The straightness shall be measured over full length of bar by mounting on three roller supports at both ends and center of each bar. The bar should touch all three sets of roller supports. This check is done by rotating the bar through 360deg on the supports.
 - DIN 1014 for Square Bars
 - a) The Standard Covers Permissible Deviation in Hot Rolled Square Bar Side length; Bar Length; Straightness, Twist, rounding off of edges
 - b) Measurements to be carried out
 - i) The side length shall be measured at the beginning & end of bar
 - ii) The Straightness & Twist has to be measured over full length of bar
- The size i.e. diameter, side length of the bar is measured using a precision vernier calipers or Micrometer of 0.01accuracy. The bar should be at room temperature at the time of measurement, and measurement is done on random sample basis in a lot.
- The tolerances are as per DIN 1013 and 1014 for which the equivalent Bureau of Indian Standard is IS1786: 1985.
- All bars not conforming to the standards mentioned above are rejected.

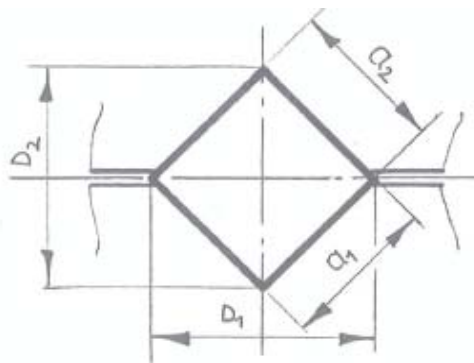
- The ovality is measured by measuring the diameter of the round bar at 0° and 90°. The bar is examined for any flattening or finning.
- A measuring tape of BIS brand, e.g. Freemans is used for measuring the straight lengths of the bars. Each bundle having short lengths beyond allowed quantity will be rejected.
- The unit weight of the bar or kg/m is measured by cutting a sample length of exactly 1m and weighing the same on an electronic lab weighing machine with error of ± 0.001 .
- Measurement points for Round Bar



- a) d_hdiameter “on high”
- b) d_gdiameter “on gap”
- c) d_{s1} , d_{s2} diameters on the “shoulders”

Every one of the 4 diameter-results must be within the tolerance-range.

- Measurement points for Square Bar



- a_1 , a_2 side lengths thickness
- D_1 , D_2 diagonals

The aim is that $a_1 = a_2$; all 4 corners/edges should be sharp and $D_1 = D_2$.

3.7.3 Physical & Chemical Properties Analysis

- For grain Structure Control of Metal of finished product i.e. Carbide-grain-size, Carbide distribution, grain-size in general, partial & total Decarburization etc on a random basis small cut pieces of bars are sent to the approved Testing Laboratory for Ultrasonic Test.
- On a random basis, bar pieces are taken to a Drill press where they are drilled and the resultant chips are sent to Testing Lab for Chemical Composition analysis to ensure that different elements like Carbon, Manganese, Sulphur, Phosphorous are within the prescribed limits as per BIS:1786 as provided at **Table 3.4**. This test is now being replaced by using a spectrometer

Table 3.4
Chemical Analysis of rolled Products as per IS 1786

Parameters (%)	Min	Max
Carbon	-----	0.30
Manganese	0.30	-----
Sulphur	-----	0.055
Phosphorus	-----	0.055

- The Tensile, Proof Stress Tests, Bend & Rebend Tests are performed on small finished bar pieces cut from bars on random basis using Universal Testing Machine. The piece is fixed in the jaws of the UTM and load applied till the Specimen fails. The readings for proof stress & Tensile Strength and percentage elongation are noted.
- In U bend test, transverse cracks should be visible if the bar is bend tested, no transverse cracks should be visible if the bar is bent by 180° around a mandrel which is 3 x dia of bar being tested. Similarly in Rebend test, no transverse cracks should be visible after bending the same through 22.5° around the same mandrel.
- The following are a list of testing instruments in a Rolling Mill:
 - Vernier calipers/ U Micrometer to measure diameter, thickness of section legs/ web flanges etc
 - High Tensile measuring 50m tape to measure the length of the bar

- Impact testing machine
- Ultimate Tensile testing machine
- Bend testing Fixture
- Spectrometer for chemical analysis

3.8 LOG BOOK FOR ROLLING MILL SUPERVISOR/INCHARGE

- A Log Book or record of all the events that have taken place on that particular day is to be maintained by the Rolling Mill Supervisor/Incharge.
- Following information need to be recorded on an hourly/daily basis:
 - Date & Time of Starting the RM in the morning
 - Grade of Input material
 - Section being rolled & sized
 - Number & Unit weight of ingots charged in RHF (total material charged)
 - Finished weight of Ingots/Billets fed to RM
 - Readings of surface temperatures of Rolling stock at end of Roughing Mill, Intermediate Mill & Finishing Mill & at entry of TMT System.
 - Lubricating Oil Temperature, Pressure, Flow Rate.
 - Cooling Water Temperature, Pressure, Flow rate
 - Weight of Misrolls & Causes, End Cuts and Scale Loss.
 - Finished Weight of Product (s).
 - Problems encountered, Down Time, Corrective action
 - Time Available for rolling, planned downtime, time used for rolling, Mill Utilisation
 - Power consumed, Specific Power Consumption.
 - Proper functioning of RM major equipments/parts.

A recommended Log Book format for RM Supervisor/Incharge is provided at **Table 3.5.**

3.9 STACKING OF FINISHED PRODUCTS & NOMENCLATURE

- After Twisting/ cooling, the Bars shall be tied into bundles manually by binding steel wire, lifted and stacked in storage godown manually. The weight of one bundle is approx 100kg. All bars of same size and treatment are stacked together.
- Bars longer than 6m are bent in the shape of an U and bundled on both legs of the U.
- Each bundle is properly tagged for identification by tying metallic Tags, giving Colour Codes at the end of the bundle and following information recorded on the tag:
 - Batch/Heat No
 - Material Grade, Size, length
 - Date of production
 - Number of Pieces
 - Weight of Bundle (kgs)
 - Name of the Party.
- The finished product will be stored as per finished length i.e. stack of 12m, 9m, 6m etc and the height of stack be restricted for ease of handling. Separators are inserted between successive layers to allow sling insertion between layers.

Table 3.5

Format of Log Book for Rolling Mill Operator

Grade of Input Material	Section being Rolled & Size	Heat No.
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Date & Time	Hour	Charged Ingots/Billets		Finished weight of Ingots/ billets fed to RM (Kg)	Surface Temperature Of Rolling Stock by using (°C) infrared pyrometer					Lubricating Oil Temperature, Pressure, Flow Rate	Cooling Water Temperature, Pressure, Flow Rate	Weight of Misrolls (Kg) & Causes	Weight of End cuts (Kg)	Finished Product Weight (Kg)	Total Scale Loss (Kg)	Yield (%)
		Number	Unit Weight (Kg)		Exit Billet/Ingot from Furnace	At end of Roughing Mill	At end of Intermediate Mill	At end of Finishing Mill	At Entry of TMT							
	1 to 2															
	2 to 3															
	3 to 4															
	4 to 5															
	5 to 6															
	6 to 7															
	7 to 8															
	8 to 9															
	9 to 10															
	10 to 11															
	11 to 12															

Date & Time	Hour	Problems Encountered	Downtime (hr)	Corrective action Taken	Planned Downtime (hr)	Time available for Rolling (hr)	Time Used for Rolling (hr)	Mill Utilization (%)	Mill Power Consumption (KWH)	Specific Power Consumption (KWH/Tonne)	Status of Major Equipments/Parts											
											Roughing Mill	Intermediate Mill	Finishing Mill	Mill Motors & Drives	Instrumentation and Control System	Repeaters	Roller Tables	Shears & Cutting Machines	TMT System	Cooling Bed	Power Supply & Distribution System	Centralized Oil & Grease Lubrication System
	1 to 2																					
	2 to 3																					
	3 to 4																					
	4 to 5																					
	5 to 6																					
	6 to 7																					
	7 to 8																					
	8 to 9																					
	9 to 10																					
	10 to 11																					
	11 to 12																					

CHAPTER 4

SAFETY ASPECTS



4.1 STANDARD SAFETY DEVICES FOR ROLLING MILL

4.1.1 Safety Guards

All moving parts have to be protected by suitable guards to prevent contact with the personnel on the shopfloor. E.g. Gearbox flywheel, couplings, V-Belt drives and spindles have to be isolated with strong guards/ cages.

4.1.2 Interlocks

Interlocks are safety devices installed in equipment to prevent accidental sequential operations which are wrong, dangerous for either equipment, product, men or all. Interlocks ensure that unless a certain condition is fulfilled further operations cannot continue.

- An important safety device in the Rolling Mills is the interlock between the tilting table on the exit side of the Rolling Mill Stand and the rotation direction of the roller table on the ingoing side. When the tilting tables on the outgoing side of the Stand is in the 'UP' position the roller table rollers on the ingoing side cannot be rotated towards the stand in order to prevent the bar from entering

the stand and hitting and damaging the tilting mechanism under the tilting table on the outgoing side.

- The next interlock is that on the end cropping shear. The end cropping shear cannot be moved to the line of rolling when there is no bar in the rolling line, otherwise the bar could hit the shear blade, damaging the blade through impact. This is actuated by a photoelectric cell with a time delay situated ahead of the shear, which senses the presence of the bar and then switches on the transverse motion of the shear, to cut a predetermined length of bar end. By adjusting the timer the length of cut can be varied as required. The control of the timer is from the pulpit operator's desk.
- If there is a malfunction at the TMT Annunciation panel, an alarm will be actuated which will blow a siren and set off flashing red lights, which can be easily heard and seen by the roughing stand pulpit operator, who will in turn alert the ejector operator to stop the ejecting of billets from the Reheating Furnace.
- The Centralized Lubricating System has the following interlocks/alarms:
 - a. Low pressure of oil – this sounds an alarm when pressure drops below 1.2 kg/cm^2 and trips the motor when pressure drops below 0.8 kg/cm^2 . The Maintenance staff resets the alarm and rectifies the pressure drop by changing over to the standby oil filter, and if situation does not improve, he switches on the standby pump. Both these actions can be done while the Mill is operating.
 - b. High temperature of oil – When the oil temperature exceeds 60°C an alarm is sounded. The Maintenance staff resets the alarm and rectifies the problem by increasing the flow of cooling water in the heat exchanger. If this does not rectify the problem the root cause is investigated by checking the bearings in the reduction gearbox and pinion gearbox by feeling the end covers by hand/ contact thermometer. If any of the bearings is found to be excessively hot, the Mill is stopped after the bars in the mill are rolled out and the end cover of the hot bearing is opened and the rollers of the bearing are checked for damage. The outer race is checked for pitting/ cracks and suitable action is taken for changing the

bearing. The oil in the filters is checked for metallic powder which could be because of damaged bearing cage/ races. The Mill is restarted by inching operation and if pressure and temperature are found OK then clearance is given to the Production staff to continue rolling.

- c. Low oil level: If the oil level in the oil reservoir tank touches the low level mark an alarm sounds out. If the oil level in running reaches 1" below the low level the Mill motor is tripped. The Maintenance staff resets the alarm and rectifies the problem by refilling the tank with oil. He investigates the cause for low level by checking and tightening all pipe joints to prevent leakages.
- d. High water temperature: If the cooling water to the heat exchanger exceeds 55⁰C an alarm is sounded. The Maintenance staff resets the alarm and rectifies the problem by checking the water temperature in the recirculation tank and if it is high then fresh water is introduced into the system. If there is a cooling tower in the system then the fan is started to cool the water
- e. Low water pressure: If the cooling water pressure drops below 1 kg/cm² is sounded. The Maintenance staff resets the alarm and rectifies the problem by checking the water pump. He can switch on the standby pump and then maintain the malfunctioning pump. He can also check the pipes for leakages and rectify the same.

It may be noted that all the above points are normally checked and action taken during the off-operating hours on daily basis. The Maintenance system itself is overhauled to prevent recurrence of such failures.

4.1.3 Alarms

Apart from the above mentioned automatic alarms there are some alarms that are initiated by the Operating staff.

- An alarm is actuated by the Roughing Stand pulpit operator when there is a cobble or misroll in the Mill, to alert all the Rolling Mills technicians on the shopfloor to help in clearing the cobble from the stand and to take corrective actions to prevent further cobbles. This alarm also alerts the ejector operator to not push any further billets into the Mill ingoing roller table until the 'ALL CLEAR' signal is given.

The EOT Crane operator reaches the site of the misroll immediately to help in clearing the cobble.

- There should be alarm buttons at every 20m on the shop floor so that anyone can press the alarm button in case he notices an abnormal and unsafe condition. All alarms should be reset only under the direction of the shift in charge.

4.1.4 Annunciations

- An annunciation Panel is provided on the TMT Cooling water tank pulpit operator's cabin. This panel clearly annunciates the bar inlet temperature, bar temperature at 10m distance from the TMT exit, water temperature, flow quantity and pressure to the various pipes and nozzles within the TMT Box.
- Temperature Readings are noted through installed Temperature Monitors/temperature gauges fitted in water cooling system; Centralised oil Lubrication & greasing system etc.
- Pressure reading are noted through installed Pressure gauges fitted in Water cooling system; Centralised Oil lubrication & greasing system; Air compressors; Hydraulic system etc.

4.1.5 Controls

- ON/OFF for Main Drive Motors through Control Panels.
- Overload Relays for Electrical Systems.
- Motor bearing temperature gauges and vibration meters are kept connected on-line to sense performance drop as soon as it occurs

4.2 SAFETY INSTRUCTIONS FOR ROLLING MILL OPERATORS

- All operators have to compulsorily be made to wear safety attire as the location and system require.
- Safety Hardware includes:
 - i. Safety overall (dungarees)

- ii. Safety Boots with steel toe caps
 - iii. Safety Boots with steel toe caps & rubber soles for electricians
 - iv. Safety Helmets
 - v. Safety Hand gloves with leather palms
 - vi. Blue goggles for welders
 - vii. Plain glass goggles to prevent sparks from hitting the eyes
 - viii. Leather elbow length hand gloves for welders
 - ix. Asbestos elbow length hand gloves for working at or near RHF
 - x. Asbestos shin protectors for working at or near RHF
 - xi. Rubber handgloves for electricians
 - xii. Safety belts for working at heights
 - xiii. Hand tool belt pouch for fitters for slide wrench, pipe spanner & allen key set
 - xiv. Hand tool belt pouch for electricians for slide wrench, screw driver cum tester
 - xv. Gumboots for working in water logged areas
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- If there is a malfunction at the TMT Annunciation panel, an alarm will be actuated which will blow a siren and set off flashing red lights, which can be easily heard and seen by the roughing stand pulpit operator, who will in turn alert the ejector operator to stop the ejecting of billets from the Reheating Furnace.
 - The ingot discharge roller table is attended by three persons. When the RM Pulpit Operator (RMPO) gives a signal of a short bell ring the ejector operator pushes out a single ingot/ billet. Normally if everything goes without problem the ingot rests on the rotating platform which is rotated by using a hook. The ingot could slip and fall towards the workers handling the table. The workers should wear safety boots with steel toecaps, heat-proof aprons, asbestos handgloves, and safety blue glasses to face the glare of the furnace heat and whiteness and also safety helmet.

- At the rotating platform, sometimes the ingot gets stuck in front of the hearth and the workers would be required to go over to the other side. Proper cross over platforms should be erected at all locations where a person is required to go to the other side of the roller table/rolling line.
- All along the length of the Rolling Mills a designated walkway about 2m wide with phosphorescent yellow lines painted on either side should be provided. All along this path there should be necessary protective handrails and expanded metal protective shield of suitable height wherever the walkway comes near the rolling line.
- The belt drives of the main mill motors should have a safety protective guard, preferably transparent acrylic. If this is not in position it should not be possible to switch ON the motor.
- All electrical panels should be provided with a rubber mat of sufficient insulation properties to prevent accidental shocks to persons working on the panel.
- No one except the EOT Crane operator or Maintenance staff should climb up the crane platform. The collector angles for the Crane should be installed on the opposite side of the operator's cabin. Otherwise the collector angles should be protected by wire mesh in the cabin with proper insulated fixing arrangement to prevent accidental touching of the power rails from the operator's cabin.
- Whenever the EOT Crane carries a load over the shopfloor where people are working the crane operator should continuously blow the siren cautioning the people down below.
- The EOT Crane hoists brakes should have two thrustors so that in case one brake fails the other acts.
- The cooling bed operators should be alerted each time the bar is rolled after a breakdown or long stoppage to be prepared for the arrival of the rolled bars which arrive at high speeds.

- While removing cobbles from Stands or Mill floors, cut them into pieces for easier, safe handling. Dragging of long cobbles on the Mill bay to be avoided to prevent damage to equipment and to persons.
- Welding jobs be properly screened to avoid eyestrain to Operators.
- Loose bolts, nuts, broken and Serviceable spare parts, grease oil, etc, should not be left on Cranes/Crane gantry after maintenance.
- Easy Access to Fire Extinguishers and Electrical Panels.

4.3 DO'S & DON'TS IN ROLLING MILL OPERATIONS

1. POWER SUPPLY & DISTRIBUTION:

- a. THE CONDUCTORS FROM THE GRID SHOULD BE SUFFICIENTLY DIMENSIONED TO TAKE CARE OF THE 300% EXPECTED OVERLOAD DRAWING.
- b. THE CABLES, TRANSFORMERS, METERING DEVICES, PROTECTION DEVICES SHOULD BE INSTALLED AS PER INDIAN ELECTRICITY RULES.
- c. THE SWITCHYARD SHOULD BE PAVED WITH STONE CHIPS & METAL, AND SHOULD BE FREE OF ANY VEGETATION.
- d. THE SWITCHYARD SHOULD BE ADEQUATELY FENCED TO PREVENT ENTRY OF UNAUTHORIZED PERSONS AND ANIMALS.
- e. THE HT POWER FACTOR IMPROVEMENT CAPACITORS SHOULD BE INSTALLED ONLY AFTER PROPER CALCULATION.
- f. TO PREVENT TRANSMISSION LOSSES IT IS NECESSARY TO TRANSMIT AT THE HIGHEST VOLTAGE POSSIBLE TO ALL AUXILIARY TRANSFORMERS, BEFORE STEPPING DOWN TO REQUIRED VOLTAGE AT THESE TRANSFORMERS CLOSE TO END USER EQUIPMENT.
- g. A CHECK METER AND POWER FACTOR INDICATOR SHOULD BE INSTALLED TO ALWAYS VERIFY THE CORRECTNESS OF THE GRID METERS.

- h. EACH AUXILIARY PANEL SHOULD BE ADEQUATELY DIMENSIONED TO WITHSTAND THE POWER & VOLTAGE IT FEEDS.**
- i. EACH AUXILIARY PANEL SHOULD BE FITTED WITH A LOCAL POWER METER AND POWER FACTOR INDICATOR TO ENABLE CONTROLS AS REQUIRED.**
- j. THE ROUTING OF THE CABLES SHOULD BE SUCH THAT NO UNPROTECTED CABLE IS LAID UNDERGROUND OR UP TO 1.5M ABOVE FLOOR LEVEL.**
- k. EACH SHED BAY SHOULD BE PROVIDED WITH ITS OWN ISOLATOR SWITCH TO ENABLE LOCALIZED MAINTENANCE.**
- l. BEFORE WORKING ON ELECTRICAL SYSTEMS, THE SYSTEM SHOULD BE ISOLATED AND SWITCHED OFF AND A DANGER TAG / 'DO NOT SWITCH ON' TAG SHOULD BE HUNG ON THE ISOLATION SWITCH.**
- m. ALL MAINTENANCE AND OPERATIONAL STAFF SHOULD TAKE A SHUTDOWN IN WRITING FROM THE ELECTRICAL STAFF, PUT THEIR OWN 'DO NOT SWITCH' TAGS APART FROM THE ELECTRICAL STAFF'S TAG ON THE ISOLATOR, BEFORE COMMENCING WORK ON ROTATING EQUIPMENT**
- n. ALL WORK MEN SHOULD BE GIVEN AN ORIENTATION TRAINING IN SAFETY SYSTEMS AND PRACTICES AND FIRE DRILLS AT THE TIME OF EMPLOYMENT AND PASS THE MINIMUM REQUIRED TESTS.**
- o. THE LOCATIONS OF THE FIRE HYDRANT POINTS, EXTINGUISHERS ETC SHOULD BE PROMINENTLY DISPLAYED AND PROTECTED FROM DAMAGE DURING PLANT OPERATIONS. SUFFICIENT HYDRANTS AND EXTINGUISHERS SHOULD BE KEPT READY FOR USE WITH PERIODICAL REFILLING AND USAGE.**
- p. PROMINENT MARKERS SHOULD BE INSTALLED AT CABLE TRENCHES/ CABLE JOINTS TO CAUTION THE WORKMEN AGAINST EXCAVATING OVER THE CABLES.**

2. ENSURE THAT ALL THE EQUIPMENT AND SYSTEMS ARE FUNCTIONAL AT THEIR DESIRED EFFICIENCY AND PARAMETERS
3. CHECK AND RECTIFY ALL MALFUNCTIONING EQUIPMENT DURING THE NIGHT/ OFF ROLLING HOURS
4. ENSURE THAT THE FACTORS AFFECTING THE COST OF PRODUCTION ARE TAKEN CARE OF LIKE SPECIFIC ELECTRICITY CONSUMPTION, SPECIFIC FUEL CONSUMPTION, YIELD, BURNING LOSSES, END CUTS AND MISROLLS OR COBBLES.
5. CHECK THE INVENTORY LEVELS OF THE VITAL OPERATIONAL CONSUMMABLES, LIKE ROLLS/ ROLLER GUIDE BOXES, ROLL & ROLLER BEARINGS, OXYGEN & ACETYLENE CYLINDERS TO ENSURE UNINTERRUPTED ROLLING.
6. STORAGE OF OXYGEN & ACETYLENE CYLINDERS HAVE TO BE IN A CORDONED OFF AREA AND AWAY FROM ACCIDENTAL SPARKS/ FLYING HOT SCALE ETC
7. ENSURE PERIODIC AND TIMELY QUALITY CHECKS, USING APPROPRIATE TEMPLATES AND MEASURING INSTRUMENTS, OF THE PRODUCT DURING THE ROLLING PROCESS, TO AVOID REJECTIONS.
8. ENSURE THAT PROPER MATCHING OF ROLLS IS DONE TO PREVENT EXCESSIVE LOOP LENGTHS OR TENSION ON BARS BETWEEN SUCCESSIVE PASSES.
9. ENSURE THAT ONLY THE CORRECT LENGTHS OF END CUTS ARE MADE TO INCREASE YIELD.
10. INSPECT AND DECIDE ON THE QUALITY ACCEPTANCE OF THE CHARGE MATERIALS BEFORE THEY ARE CHARGED INTO THE REHEATING FURNACE TO AVOID PROBLEMS IN SMOOTH ROLLING.
11. DO NOT CONTINUE ROLLING WHEN A MISROLL HAPPENS UNTIL THE PROBLEM IS RESOLVED.
12. THE MISROLLED BAR SHOULD BE CLEARED IMMEDIATELY AND SHIFTED OUTSIDE THE SHOPFLOOR SO THAT NO OBSTRUCTION IS THERE ON THE SHOPFLOOR WHICH ARE POTENTIAL HAZARDS.

13. ENSURE THAT THE LUBRICATION EQUIPMENT ARE FUNCTIONING PROPERLY AND THAT PROPER LEVELS OF LUBRICANTS ARE MAINTAINED IN THE RESERVOIRS/ TANKS.
14. ENSURE THAT ALL THE ROLL BEARINGS ARE PROPERLY LUBRICATED BEFORE COMMENCEMENT OF ROLLING.
15. ENSURE THAT PROPER QUALITY OF COOLING WATER AT THE PROPER PRESSURE & TEMPERATURE IS AVAILABLE AT THE USAGE POINTS ESPECIALLY FOR ROLL COOLING AND TMT BOXES.
16. IN CASE OF ACCIDENTS:
 - a) EACH DEPARTMENT SHOULD HAVE TRAINED AND QUALIFIED FIRST AID ADMINISTRATION STAFF. IN CASE OF ACCIDENTS THESE FIRST AID WORKERS SHOULD ADMINISTER FIRST AID AS REQUIRED AND INSTRUCT THE SHIFT-IN-CHARGE TO CALL FOR THE AMBULANCE TO TRANSFER THE INJURED PERSON TO THE NEAREST DISPENSARY, FROM WHERE THE DOCTOR WILL INSTRUCT ON FURTHER ACTION TO BE TAKEN.
 - b) THE SHIFT IN CHARGE WILL HAVE TO FILL IN THE ACCIDENT FORMS GIVING DETAILS AS TO TIME OF ACCIDENT OCCURRENCE, BRIEF DESCRIPTION AS TO HOW THE ACCIDENT HAPPENED. THE FORM WILL HAVE TO BE SIGNED BY TWO WITNESSES AND TWO COPIES WILL HAVE TO BE HANDED OVER TO THE DOCTOR AND ONE COPY KEPT IN THE FILE. HE WILL IMMEDIATELY INFORM HIS SUPERVISOR WHO WILL INFORM THE GENERAL MANAGER. THE GENERAL MANAGER WILL AS SOON AS POSSIBLE COME TO THE SHOP-FLOOR AND CONDUCT AN ON -THE-SPOT INQUIRY AND MAKE A REPORT AND GIVE APPROPRIATE INSTRUCTIONS TO THE WORKERS TO PREVENT A RECURRENCE OF THE ACCIDENT.
 - c) IF THE ACCIDENT IS FATAL THEN A POLICE CASE RESULTS LEADING TO INVESTIGATION BY BOTH THE POLICE AND THE FACTORY INSPECTOR TO ESTABLISH THE CAUSE OF ACCIDENT AND PREVENTIVE MEASURES TO BE INCORPORATED. IN CASE IT IS ESTABLISHED THAT THE ACCIDENT WAS DUE TO NEGLIGENCE LIKE ABSENCE OF SAFETY WEAR/ GUARDS ETC THEN THE CEO IS HELD LIABLE AND A CASE IS INSTITUTED AGAINST HIM.